

Attachment M

Design and Engineering Calculations of CWIS

Luminant Generation Company LLC
Trinidad Steam Electric Station
TPDES Permit No. WQ0000947000

40 CFR 122.21 IMPACTED BY 316(b) of Clean Water Act
Trinidad Steam Electric Station
Cooling Water Intake

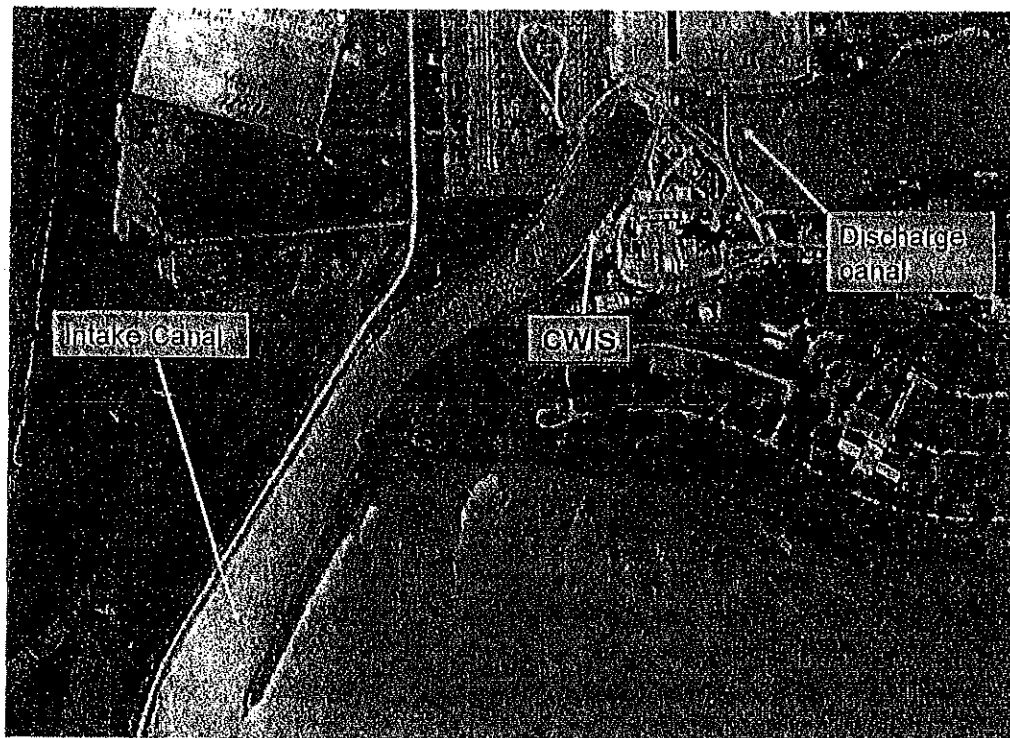
EPA signed into regulation new requirements for existing electric power generating facilities for compliance with Section 316(b) of the Clean Water Act on July 9, 2004. These regulations became effective on September 7, 2004 and are based on numeric performance standards. The Rule at 40 CFR 125 impacts 40 CFR 122.21 (Application for a permit) by requiring existing facilities defined in Part 125 to submit for review the following information:

- (1) A narrative description of the operation of the cooling water system, its relationship to cooling water intake structures, the proportion of the design intake flow that is used in the system, the number of days of the year the cooling water system is in operation and seasonal changes in the operation of the system, if applicable;

Trinidad Steam Electric Station (TRSES) is located on the northern design shoreline of Trinidad Reservoir in Henderson County, TX. Trinidad is an off-channel man-made reservoir, created in 1926, and is owned and operated by TXU. The following figure provides an aerial photograph showing Trinidad's intake on the reservoir. At the conservation pool level of 283 ft. msl, the reservoir has a surface area of 1080 acres. TRSES is a single-unit natural gas-fired facility rated at 240 MW utilizing a once-through cooling water system. TRSES is designated for seasonal operation, and its 2004 capacity factor was about 2.5%.

TRSES has one cooling water intake structure (CWIS), located on the east side of the canal, approximately 1,700 feet from the mouth of the 1,900-foot long intake canal. The intake canal originally supported other generating Units that have since been retired. The canal is approximately 150 feet wide in front of the CWIS. The CWIS has three bays, each equipped with one vertical, mixed-flow circulating water pump, located downstream of traveling water screens. Bays 1 and 2 also are equipped with small fire water pumps. The three circulating water pumps have a rated capacity of 130.3 cfps (66,000 gpm) each. When all three circulating water pumps are operating, the flow to the facility is 441.1 cfps (198,000 gpm). There is a trash rack located at the front of each bay that prevents large debris from reaching the traveling water screens that are located about 13 feet downstream of the trash racks. The screens for each CWIS are 10-feet wide and have a 3/8-in. square mesh. This intake structure was designed with a raised threshold to each bay. The height of the threshold is EL 275 ft, which is above the bay bottom of EL 269.9 ft.

Velocities within the CWIS bays were calculated at the normal operating pool water level (El. 285 ft.) and at the low water level (EL. 280 ft.) Because the smallest cross-sectional area of each bay is found at the threshold, the maximum water velocity will occur at that location in the bay. At normal operating water level, the velocity in each bay is 1.3 fps. At minimum operating water level the velocity in each bay is 2.7 fps.



Trinidad Steam Electric Station (TRSES)

- (2) Design and engineering calculations prepared by a qualified professional and supporting data to support the description listed above.

See the following documentation.

PALCO Engineering & Construction Services



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VELOCITY CALCULATIONS FOR TRINIDAD SES UNIT 6

Trinidad SES Intake Structure contains three (3) intake bays.

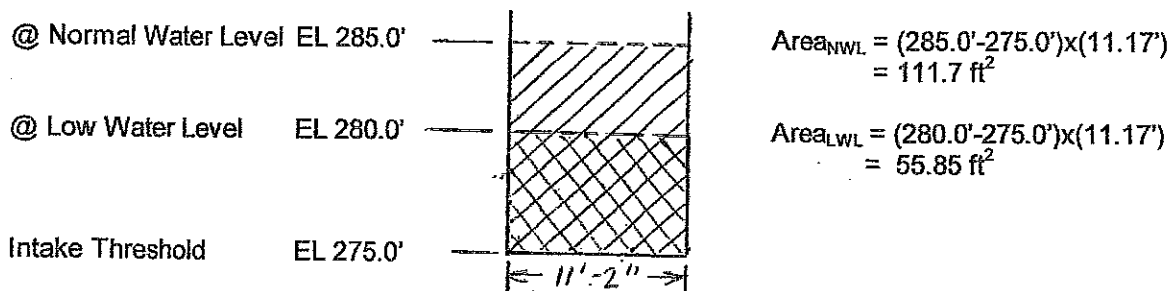
Bays 1 and 2 each contain:

1 Circulating Water Pump	66,000 gpm	(ref. Curve HTA-25491-A)
1 Fire Water Pump	750 gpm	(ref. Curve 88-02132-1A)

Bay 3 contains:

1 Circulating Water Pump	66,000 gpm	(ref. Curve HTA-25491-A)
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Each Bay has the same dimensions and thus the same cross-section flow area (ref. Dwgs. C-632A and C-628C for dimensions and water levels). This intake structure is designed with a threshold at El. 275'-0" that is above the bay bottom of El. 269'-11". This threshold reduces the cross-sectional area making the water velocity greatest at that location, not at the traveling water screen.



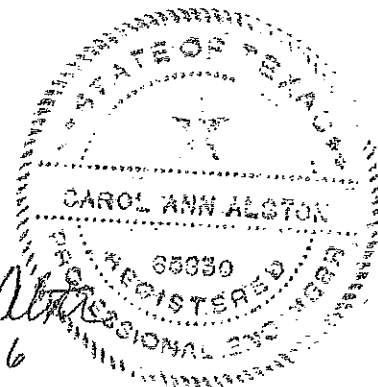
Maximum Water Velocity is equal for both Bays 1 and 2 and will occur when both the circulating water and fire water pumps are running:

$$\text{Velocity} = \frac{\text{Flow Rate}}{\text{Flow Area}}$$

$$\text{Velocity @ Low Water Level} = \frac{(66,000 + 750) \text{ gal}}{\text{min}} \times \frac{1 \text{ ft}^3}{7.48 \text{ gal}} \times \frac{\text{min}}{60 \text{ s}} = 2.66 \text{ ft/s}$$

$$\text{Velocity @ Normal Water Level} = \frac{(66,000 + 750) \text{ gal}}{\text{min}} \times \frac{1 \text{ ft}^3}{7.48 \text{ gal}} \times \frac{\text{min}}{60 \text{ s}} = 1.33 \text{ ft/s}$$

Carol Ann Alston
5/25/06

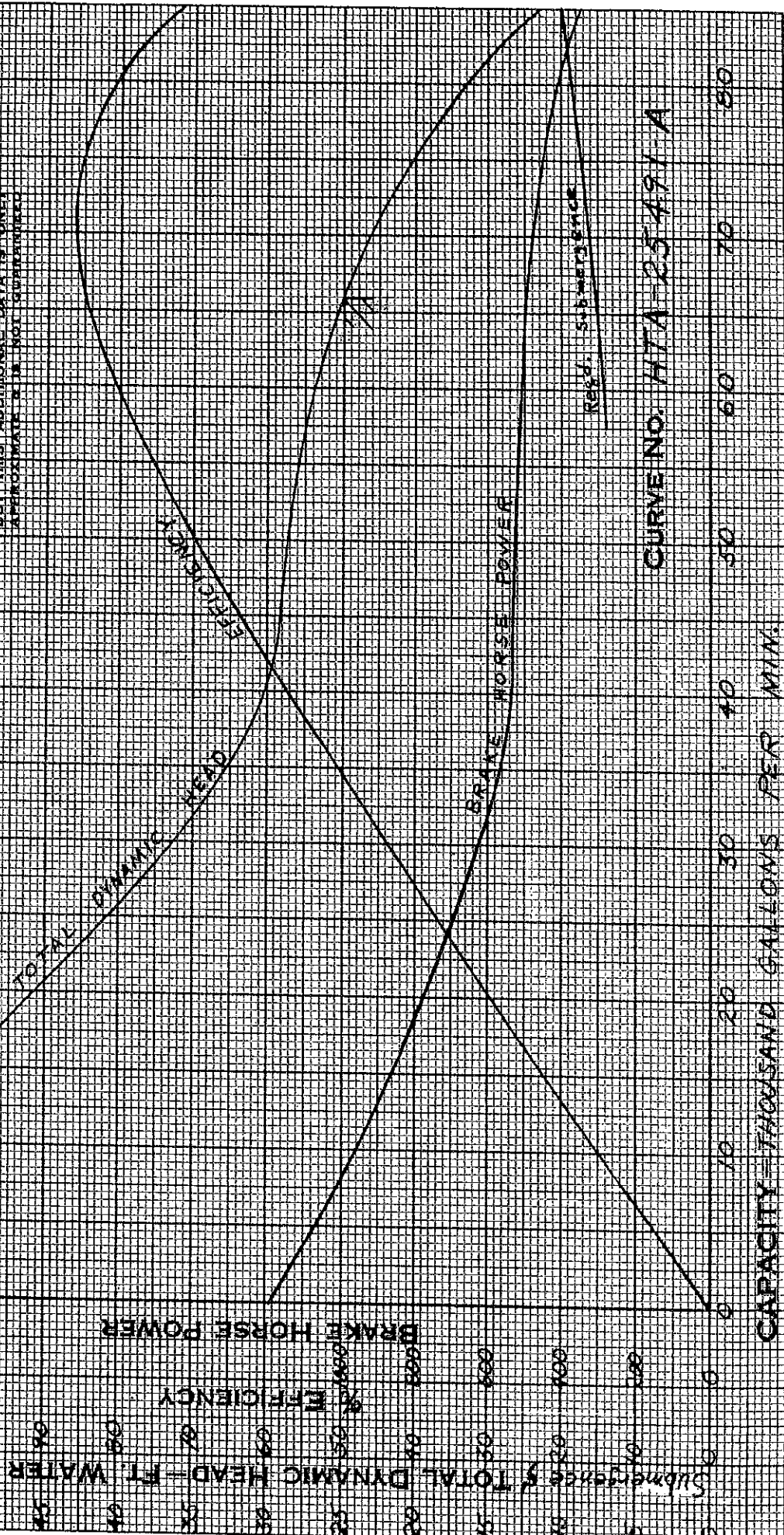


WESTINGHOUSE ELECTRIC CORP.
54 GN CIRCULATING PUMP

THIS PUMP IS GUARANTEED TO DELIVER
2500 G.P.M. 100 FEET TOTAL DYNAMIC HEAD
AT 1750 R.P.M.
AT 85% EFFICIENCY (MOTOR LOSSES NOT INCLUDED)

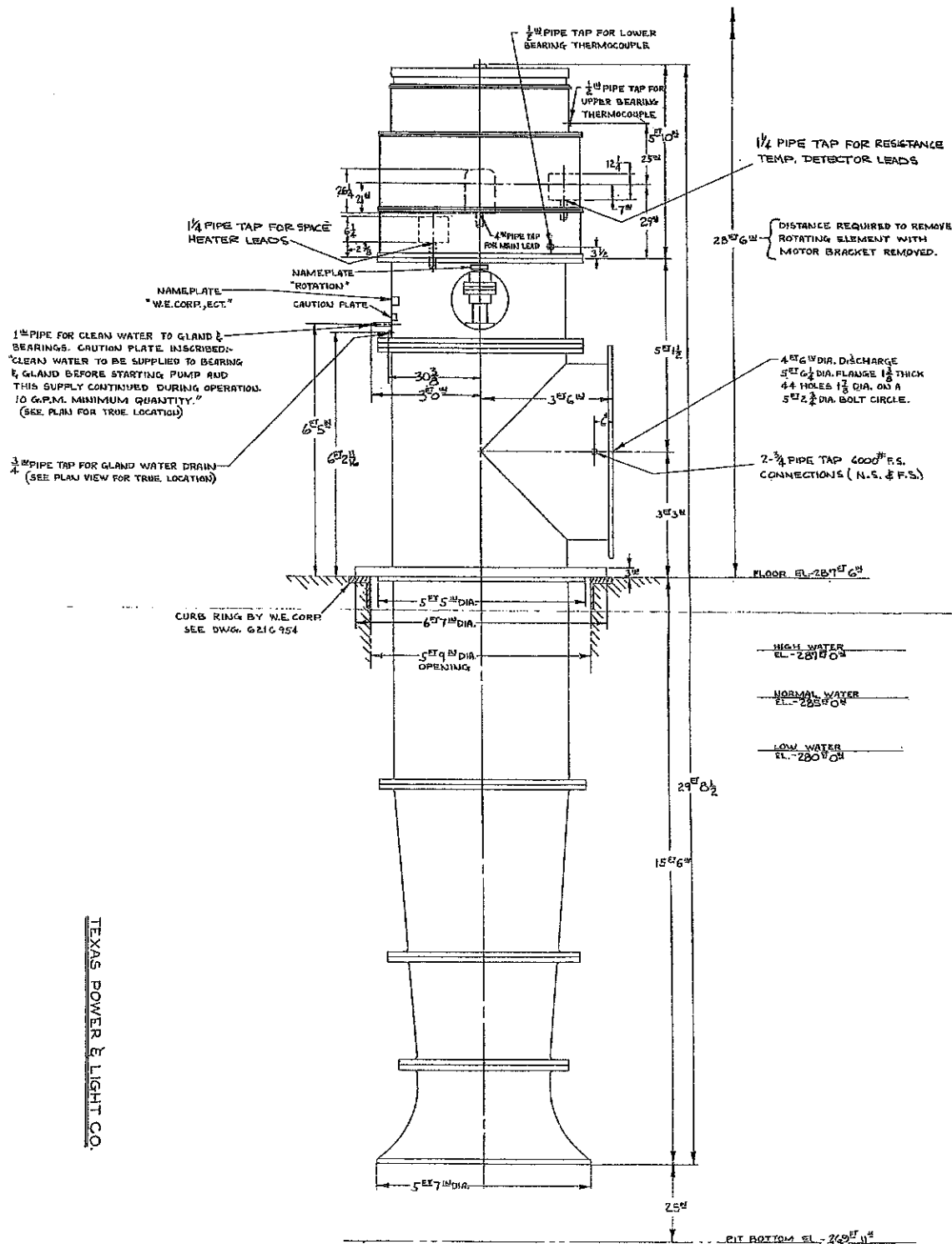
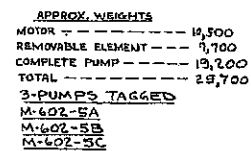
85% EFFICIENCY (MOTOR LOSSES NOT INCLUDED)

1750 R.P.M.
100 FEET DYNAMIC HEAD
THIS CURVE SHOWS EXPECTED HEAD
& EFFICIENCY AT OTHER CAPACITIES
BUT THIS ADDITIONAL DATA IS ONLY
APPROXIMATE & IS NOT GUARANTEED



CURVE NO. HTA-25491-A

CAPACITY—THOUSAND GALLONS PER MIN.



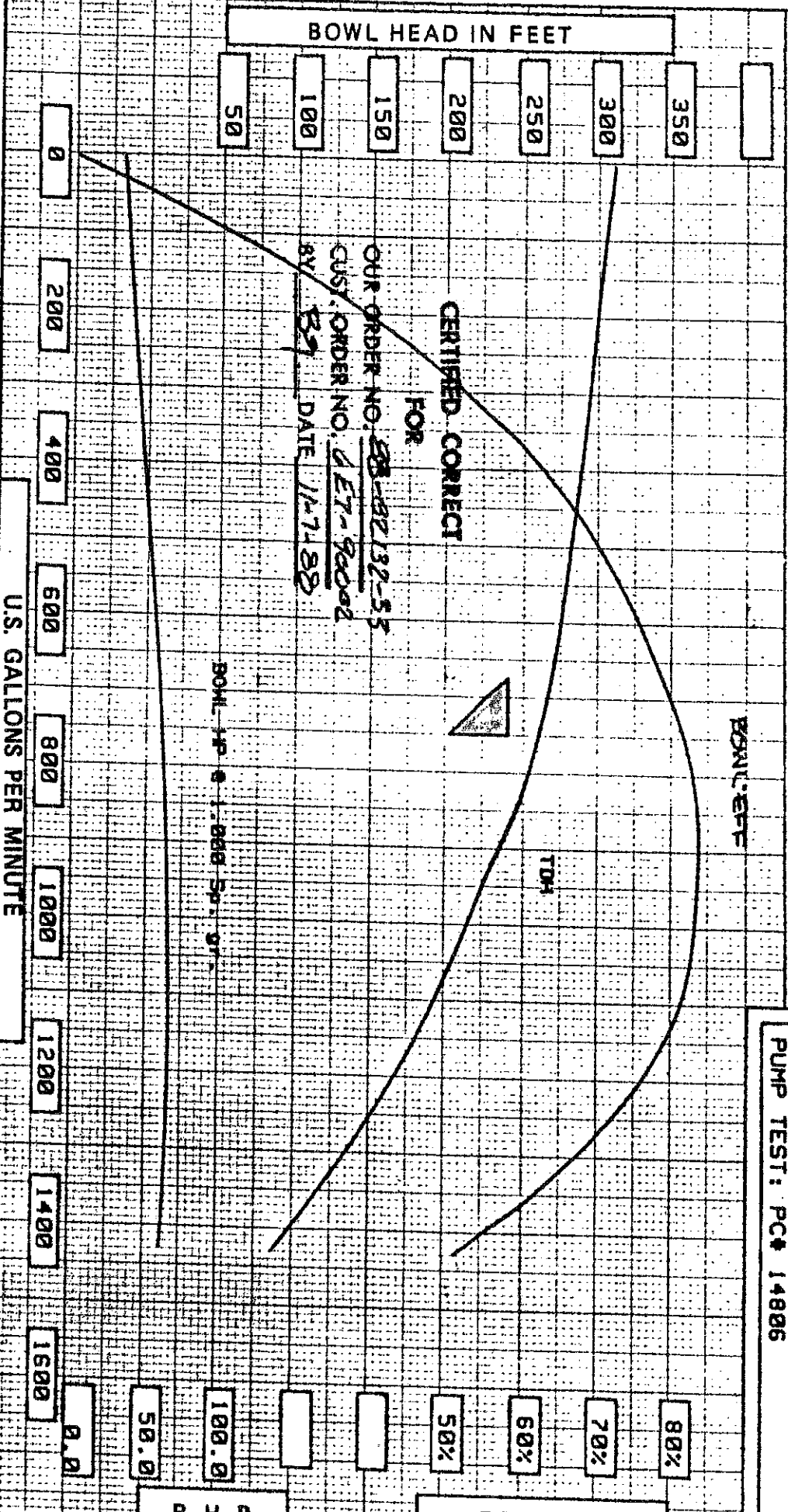
I. L. 1310-1061 – FIGURE 1 – 54 BN Circulating Pump Outline

FRAME 54 BN CIRCULATING PUMP
TITLE OUTLINE

TEXAS POWER & LIGHT CO

Performance shown is the result of a test conducted in accordance with latest Hydraulic Institute Standards. Other Standards applicable when mutually agreed upon.

TEXAS UTILITIES ELECTRIC CO.
TRINIDAD STEAM ELECTRIC STATION
TRINIDAD TEXAS
750 GPM UL/FM LABELED FIRE PUMP
PUMP TEST: PC# 14806



Peabody Floway

TYPE 12LKM-FP
NO. OF STAGES 4
R.P.M. 1725
PUMP SERIAL NO. 88-02132

DWG. NO. 88-02132-1A

"There's a Difference"

OWN. BY B.M.G. DATE 1/7/88

Attachment N

**Supplemental Fisheries Data Analysis
for Proposal for Information
Collection, Clean Water Act,
Section 316(b) Phase II Requirements**

Luminant Generation Company LLC
Trinidad Steam Electric Station
TPDES Permit No. WQ0000947000

Document No. 050019
PBS&J Job No. 441482

**SUPPLEMENTAL FISHERIES DATA ANALYSIS
FOR PROPOSAL FOR INFORMATION
COLLECTION, CLEAN WATER ACT,
SECTION 316(B) PHASE II REQUIREMENTS**

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January 2005

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Acronyms and Abbreviations

ANOVA	analysis of variance
CDS	Comprehensive Demonstration Study
CPUE	catch-per-unit-effort
CWIS	cooling-water intake structure
EPA	U.S. Environmental Protection Agency
FWS	U.S. Fish and Wildlife Service
I&E	Impingement and Entrainment
mgd	million gallons per day
PIC	Proposal for Information Collection
SDR	Shoreline Development Ratio
TCEQ	Texas Commission on Environmental Quality
TPDES	Texas Pollutant Discharge Elimination System
TPWD	Texas Parks and Wildlife Department
TXU	Texas Utilities Generating

1.0 BACKGROUND

The Clean Water Act Section 316(b) pertains to Impingement and Entrainment (I&E) of aquatic life at power-generating facilities that withdraw surface water from waters of the U.S. for cooling purposes. The U.S. Environmental Protection Agency (EPA) developed a three-phased approach for rule implementation. Phase I went into effect on January 17, 2002, and regulates new facilities. Phase II regulates existing facilities that withdraw 50 million gallons of water per day (mgd) or more. These regulations went into effect on September 9, 2004. Phase III rules are currently being developed. In Texas, many power-generating facilities withdraw water from reservoirs and are regulated under the Phase II rules. Facilities located on reservoirs are required to meet a performance standard of 80 to 95% reduction in impingement mortality (40 CFR 125.94(b)(1)). The rules provide options for meeting compliance standards and are listed in 40 CFR 125.94(a).

These new rules have been implemented through the Texas Pollutant Discharge Elimination System (TPDES). Through TPDES, many Texas facilities regulated by Phase II rules will be required to submit a Comprehensive Demonstration Study (CDS) (40 CFR 125.95(b)) by January 2008. The purpose of the CDS for facilities located on reservoirs is to characterize impingement mortality, describe the operation of the cooling-water intake structure (CWIS), and confirm that the technologies, operational measures, and/or restoration measures selected are meeting or will meet impingement performance standards. One of the first steps in developing a CDS is to prepare and submit a Proposal for Information Collection (PIC) to the Texas Commission on Environmental Quality (TCEQ). Specific components of the PIC are listed in 40 CFR 125.95 (b)(1).

1.1 PHASE II AND TEXAS UTILITIES

Texas Utilities Generating (TXU) owns and operated 18 facilities in Texas that will be regulated by the Phase II rules. All of these facilities are located on reservoirs, most of which were constructed specifically to provide cooling water for "once-through" circulation. All of the source waters for the facilities are classified as waters of the U.S. and will be subject to the Phase II requirements for lakes and reservoirs. These facilities and associated reservoir names are listed below:

Facility Name	Reservoir Name	Stream Channel	Public Access
Hubbard	Ray Hubbard Reservoir	Mainstem	yes
Stryker	Striker Reservoir	Off-channel	yes
Twin Oak		Off-channel	no
Monticello	Monticello Reservoir	Off-channel	yes
Forest Grove		Off-channel	no
De Cordova	Granbury Reservoir	Mainstem	yes
Big Brown	Fairfield Reservoir	Off-channel	yes
Comanche Peak	Squaw Creek Reservoir	Off-channel	no

Facility Name	Reservoir Name	Stream Channel	Public Access
Lake Creek	Lake Creek Reservoir	Off-channel	no
Morgan Creek	Colorado City Reservoir	Off-channel	yes
Martin Creek	Martin Creek Reservoir	Off-channel	yes
North Lake	North Reservoir	Off-channel	no
Tradinghouse	Tradinghouse Creek Reservoir	Off-channel	yes
Trinidad	Trinidad Reservoir	Off-channel	no
Valley Lake	Valley Lake Reservoir (Brushy Creek)	Off-channel	no
Eagle Mountain	Eagle Mountain Reservoir	Mainstem	yes
Graham	Graham Reservoir	Off-channel	yes
River Crest*	River Crest Reservoir	Off-channel	No

*Announced for retirement and will not be considered for 316(b) compliance.

Mainstem – Impoundment of a major river or stream.

Off-channel – Perched reservoir or impoundment of a small stream.

TXU is currently investigating a feasible and practical approach for bringing each of its facilities into compliance under these new rules. Of particular interest is the development of an approach for characterizing aquatic life and monitoring impingement at such a large set of facilities. With the exception of Comanche Peak (Squaw Creek Reservoir), there are little to no recent impingement monitoring data with which to develop required information. Conducting impingement monitoring (and possibly mortality studies) at all of these facilities would be expensive and would present significant logistical challenges. Under the existing deregulated electric supply market, it is difficult and sometimes impossible to know when or whether a facility will be in operation. Some of the smaller facilities have a very low capacity factor, or are inactive, and others have never been completed, but all maintain TPDES permits. Thus, developing a sampling program for every facility would be impractical. On the other hand, some of TXU's facilities are considered "base-loaded" and operate almost continuously, providing good opportunities to conduct thorough impingement demonstration studies. As discussed in the following sections and in the PIC document, these facilities, however, share many similarities, which offer the opportunity to consolidate some of the required data collection for the purpose of estimating facility impacts. With the range of operational scenarios presented by these facilities, TXU proposes to conduct impingement demonstrations at three of its base-loaded facilities: Big Brown (Fairfield Reservoir), Comanche Peak (Squaw Creek Reservoir), and Monticello (Monticello Reservoir). These three facilities would be considered "representative" facilities in which detailed impingement monitoring would be conducted. Data from these facilities and reservoirs would be extrapolated to represent impingement mortality at the remainder of the facilities given that they share acceptable facility, operational, and biological similarities. The purpose of this document is to evaluate and compare available fisheries and habitat data for each reservoir and to determine whether similarities exist between the representative reservoirs and the remainder of the reservoirs. Reservoirs that share similarities (if any) with the representative reservoirs will be grouped accordingly. Information gained from this evaluation will be used as a supplement to the PIC for the purpose of developing impingement demonstration strategies.

TXU facilities are sited throughout east, north, and central Texas and are located in a variety of ecological regions of Texas, as described by Hubbs (1982). These include the East Texas timber country, blackland prairies, cross timbers/grand prairies, and the plains. According to Hubbs et al. (1991), over 150 fish species occur throughout these regions, collectively. Since only one large natural lake occurs in Texas, most of these species are adapted to lotic (stream) systems. Construction of reservoirs on perennial streams alters fish communities through chemical, biological, and physical changes in these systems (Yeager, 1993). Thus, fish communities in reservoirs largely deviate from communities originally found in the pre-impounded reach. Since most large reservoirs across the state are managed for fishing, the introduction of various species for sport-fish management amplifies these changes. A number of tools exist for measuring the health or integrity of natural stream-fish communities. One example is the widely applied Index of Biological Integrity developed by Karr et al. (1986). Because of the "artificial" nature of reservoirs, few studies have focused on reservoirs in this context, which is an important factor when considering data sources for reservoirs (discussed in more detail in Section 2.0). The artificial nature of reservoirs was, in part, the logic the EPA applied when developing the tiered approach for waterbody sensitivity, whereby lakes and reservoirs were considered the least sensitive of the ecosystems categories (EPA, 2004).

Contrary to the above, reservoirs constructed "off channel" or on small, or in some cases ephemeral streams, provide habitat for fish that did not exist in the pre-impounded area. This is particularly true for many off-channel cooling ponds, such as some of the TXU reservoirs. Construction of these facilities and reservoirs created large areas of aquatic habitats, which, in turn, provided the opportunity for Texas Parks and Wildlife Department (TPWD) to stock and manage fish populations for public fishing opportunities. TXU facilities that are currently closed to the public are closed due to security and/or safety concerns, or in some cases, the facilities (and reservoirs) were never completed.

2.0 DATA SOURCES

TPWD maintains fisheries data for the TXU reservoirs open or previously open to the public. Facilities that are no longer open to the public are not currently surveyed by TPWD. Surveys are generally conducted about once every 3 years. In some cases, surveys were conducted at higher frequencies. With the exception of habitat observations, these surveys are standardized between survey years and across reservoirs and fisheries districts. The surveys include boat electrofishing, gill netting, trap netting, and periodic measurements or observations of fish habitat. Each survey takes place within a period of 1 year and each survey technique is typically employed once per survey period. The data are summarized in performance reports required by the Federal Aid in Fisheries Restoration Act. Most of the performance reports contain data summaries from previous surveys. For this study, all performance reports that represent about the last 15 years of data were requested from TPWD. The table below lists all of the survey years that were provided in the performance reports. If a performance report was not provided, it was concluded that data were not available for that reservoir. References for the performance reports used in this study are in Section 5.0.

Reservoir	Performance Report Year(s)
Colorado City	1985, 1988, 1991, 1994, 1995, 1997, 2001
Eagle Mountain	1988, 1989, 1990, 1991, 1992, 1995, 1997, 2000
Graham	1983, 1986, 1990, 1994, 1997, 2001
Granbury	1981, 1988, 1990, 1992, 1995, 1998, 2001
Fairfield	1985, 1987, 1988, 1991, 1993, 1996, 1999, 2000
Forest Grove	ND
Lake Creek	1984
Martin Creek	1985, 1991, 1994, 1999, 2000, 2001
Monticello	1987, 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999
North	1988, 1991, 1992, 1995, 1998,
Ray Hubbard	1989, 1991, 1994, 1997, 2000,
Squaw Creek	1981, 1988, 1990, 1994, 1997,
Striker	1975, 1988, 1991, 1992, 1993, 1995, 1997, 1999, 2003
Tradinghouse Creek	1987, 1989, 1991, 1994, 1997, 2000
Trinidad	ND
Twin Oak	ND
Valley	ND

ND = Data not available

In addition, the latest reports containing standing crop (cove rotenone) data were also requested. For some reservoirs, creel (angler) surveys were provided, but surveys were available for only a limited number of reservoirs and, therefore, were not considered for this study. A search of the University of Texas Library

and TCEQ archives was conducted to obtain any additional fisheries studies to complement this data set, but none were located.

3.0 DATA HISTORY AND ANALYSIS

Recent data provided for this study allows for some level of spatial and temporal analysis. Multiple data points from the last 15 years or so of data collection were used to develop reasonable statistical analysis. However, there are important factors regarding how these data were collected and the reason other types of data are not available, most of which pertain to the evolution and purposes of fisheries sampling in Texas reservoirs. When evaluating data sets that represent multiple years, or data sets that represent different reservoirs, it is tempting to assume that one data set is directly comparable with the next. This is not necessarily the case due to any number of variables, including constantly changing environmental conditions (i.e., weather, fluctuating water levels, and habitat availability), small sample sizes, stocking, occurrence of invasive species, and sample quality associated with constantly improving sampling techniques. The following section provides a discussion on the later, which is a brief history of fishery sampling in Texas public waters.

3.1 DATA HISTORY

Routine fisheries, water quality, and habitat data have been collected for Texas reservoirs since at least the 1960s. Data collection up to about the mid 1980s was quite different than current data collection efforts. During this period, rotenone, seines, and multifilament gill nets were standard sampling tools and a wide suite of water quality parameters were also measured including temperature and dissolved oxygen profiles. Current data collection relies on electrofishing, gill netting, and trap netting. The following paragraphs provide a brief history on each of these sampling techniques.

Cove rotenone – Cove rotenone sampling involved blocking coves of known areas and volumes with block nets and then applying rotenone (fish toxicant) to the sample area. Most of the dead fish eventually floated to the surface and were collected. Data collection included species identification, enumerating, measuring, and weighing the fish. This provided an estimate of standing crop (pounds/acre), numbers of fish per given area, and various population indices. These data were often extrapolated to provide estimates for entire reservoirs or to help assess angler harvest. This technique was quite effective for sampling fish that occur in shallow coves. However, it provided a skewed view of the reservoir standing crop and population composition and structure simply because the fish that occurred in the coves did not necessarily represent fish found in open waters (Nielsen and Johnson, 1985). Due to public concern over killing large numbers of fish and the evolution of alternative, less damaging sampling techniques, the use of cove rotenone surveys ceased by the late 1980s. Because cove rotenone surveys, although quite dated, are some of the only data that provide quantitative assessments of fish populations, the latest (most recent) cove rotenone data were obtained (if available) for each reservoir for this study. With some broad assumptions and validation with recent data, the standing crop estimates might provide one tool for estimating impingement rates.

Seine surveys – Seine surveys of shoreline zones were also a standard tool during this period. Seine hauls were made over a known area to provide quantitative estimates of smaller individuals and species that do not attain larger sizes. Similar to cove rotenone data, seine data can provide useful information about the segment of the population subject to impingement. However, seine surveys were discontinued for most reservoirs by the late 1980s. Due to the age and scarcity of these data, seine data were not considered for this study.

Gill netting – Gill netting is a passive sampling tool that has been applied since monitoring began. However, during the late 1980s, the types of gill nets changed. Formerly, gill nets were constructed from multifilament nylon string. Gill nets used today are monofilament, which is considered more effective at capturing fish. Also, during the early 1990s, gill net lengths were slightly shortened. Gill nets are currently constructed with various sizes of mesh within each net and are used to target a variety of fish sizes. Gill nets generally target larger bottom-dwelling and open-water species, most of which are not prone to impingement due to their large size. However, these data provide useful qualitative information about species composition and relative abundance. Gill net data for this study include only data collected using the current gill net configuration. Catch-per-unit-effort (CPUE) (net-night) refers to the number of fish caught in one gill net over about a 24-hour period.

Trap netting – Trap netting is a passive sampling technique used to target sub-adult and adult crappie (*Pomoxis* spp.). TPWD began using trap nets as a standard sampling tool in the early 1990s. Trap net data prior to the 1990s essentially does not exist. Similar to gill nets, trap nets are set out for about a 24-hour period. CPUE refers to the number of fish caught during one net-night. All of the trap net data for the period of records were obtained for this study.

Electrofishing – Boat electrofishing is an effective tool for sampling shallow, shoreline areas. Electrofishing data are largely qualitative, but provide useful information regarding species composition, relative abundance, and individual-specific indices, and can also demonstrate population trends over time. Electrofishing, in turn, can provide information that might be useful for projecting impingement rates, particularly for the sunfish (*Lepomis* and *Micropterus* spp.).

Boat electrofishing technology has greatly improved over time. TPWD began using boat electrofishing in the mid 1980s. During this period, the boats were generally small, generators were low wattage and only supplied alternating current, electrode specifications were widely variable, and frequencies were not adjustable. As such, catch rates were relatively low and varied from fisheries district to fisheries district, depending upon the type of electrofishing boat employed. Electrofishing technology rapidly improved, and by the early 1990s TPWD acquired larger, more efficient boats, which were similar between fisheries districts. As a result, electrofishing catch rates increased and data were more comparable between reservoirs. Prior to the mid to late 1990s, electrofishing sampling was conducted at fixed stations. This meant that the stations were selected by the biologists and the same stations were sampled from one survey period to the next. Each station was surveyed for 15 minutes. Existing electrofishing protocol

requires selecting random sampling stations at a higher frequency (number of stations) with a shorter (5-minute) duration. There has been some debate regarding the compatibility of the fixed station and random station data. Most likely, the degree of the impact of changing the sampling protocol probably varies from reservoir to reservoir. The potential spatial and temporal variations should be kept in mind for data evaluation. CPUE for electrofishing is expressed in terms of the number of fish captured during 1 hour of electrofishing. Electrofishing data for the period of record (dating back to the mid 1980s) were obtained for this study.

Habitat – Comprehensive water quality and reservoir profile surveys were conducted up until about the mid 1980s and those data were provided in the earlier performance reports. Existing water quality measurements include surface reading of the basic water quality parameters (temperature, dissolved oxygen, conductivity, and Secchi depth), which are usually taken at each sample station. Approximate conductivities and Secchi depths are listed in Table 1.

Availability (or lack of) of physical habitats play an important role in shaping fish communities (Miller et al., 1987). Variation in habitat might, in part, help to explain variation in fish populations. Physical fish habitats have been documented over the years by TPWD and can be found in most of the performance reports. Habitats can vary widely from year to year due to water level fluctuations, urban development, and the natural and man-induced dynamics of aquatic vegetation coverage. For most of the TXU reservoirs, water levels are maintained within a narrow range of elevations, which helps to reduce the variability induced by changing water levels. Exceptions to this include the reservoirs in the central and western part of the state. TPWD efforts for standardizing habitat data collection have been somewhat hindered due to this variability and the numerous potential combinations of habitat types. Nevertheless, habitats might provide important insight for comparative analysis. The most recent habitat survey for each reservoir was obtained for this study.

3.2 DATA ANALYSIS

Data from the performance reports provided by TPWD were entered into Microsoft Excel spreadsheets. Data fields included reservoir physical characteristics (Table 1), abundance or CPUE of selected species (Table 2), cove rotenone results (Table 3), and habitats surveyed (Table 4). Some winnowing of the fisheries data were required to develop a suite of species that are common across reservoirs and are relevant when assessing impingement. Not all species or sizes of fish are impinged. Since EPA (2004) established 3/8-inch mesh screen as one of the baseline conditions, many smaller species such as silversides (*Menidia* spp.), minnows (*Notropis* and *Cyprinella* spp.), mosquitofish (*Gambusia affinis*), and darters (*Etheostom* and *Percina* spp.) may simply pass through intake screens and do not show up in impingement samples in appreciable numbers. Unlike studies of streams that focus on species diversity, few data are available for non-game species in Texas reservoirs. Survey techniques currently employed by TPWD are biased towards larger species. Some non-game species are collected and reported by TPWD, but their data collection efforts focus on sport fish and the forage species that support sport fish.

Therefore, data on non-game species vary widely and preclude population-level assessments of these species.

Adults of larger species can typically overcome water currents and are usually not impinged. However, catch rates of larger species can provide some insight into potential impingement of juveniles of the same species. In general, the species most prone to impingement in Texas reservoirs are shad (*Dorosoma* spp.) and sunfish (*Lepomis* spp.). These species have high reproductive rates, are generally abundant, and remain of size susceptible to impingement for a large portion of their life.

This study focused on the species most susceptible to impingement and/or common sport fish and forage species. While some of the data were viewed on a species level, the statistical analysis was conducted with groups of like species. This was done to account for spatial variability in species composition of closely related species that occupy comparable niches. These groups are collectively known as the "indicator" species. This grouping generally corresponds to the genus level and are classified as follows:

Species	Grouping
Largemouth bass (<i>Micropterus salmoides</i>)	Black bass
Spotted bass (<i>M. punctulatus</i>)	Black bass
Glizzard shad (<i>Dorosoma cepedianum</i>)	Shad
Threadfin shad (<i>D. petenense</i>)	Shad
Bluegill (<i>Lepomis macrochirus</i>)	Sunfish
Redear sunfish (<i>L. microlophus</i>)	Sunfish
Longear sunfish (<i>L. megalotus</i>)	Sunfish
Warmouth (<i>L. gulosus</i>)	Sunfish
Green sunfish (<i>L. cyanellus</i>)	Sunfish
Orangespotted sunfish (<i>L. humilis</i>)	Sunfish
White crappie (<i>Pomoxis annularis</i>)	Crappie
Black crappie (<i>P. nigromaculatus</i>)	Crappie
Channel catfish (<i>Ictalurus punctatus</i>)	Catfish
Blue catfish (<i>I. furcatus</i>)	Catfish
White bass (<i>Morone chrysops</i>)	Temperate bass
Striped bass (<i>M. saxatilis</i>)	Temperate bass
Palmetto bass (<i>M. chrysops</i> x <i>M. saxatilis</i>)	Temperate bass

3.2.1 Indicator Species Abundance

Temperate bass, crappie, and catfish numbers were low when compared with the number of individuals captured by electrofishing (see Table 2). This is simply due to the sampling techniques employed by TPWD for these species. Temperate bass and catfish are sampled with gill nets and crappie are sampled with trap nets. These sampling tools are passive capture techniques that target larger, sub-adult and adult individuals (Nielson and Johnson, 1986). Numbers of individuals usually decrease with increasing age

class. It is common for most of the individuals in a population to occur within the age-0 class (first year of life), which is the segment of the population that is generally missed by these sampling techniques. Conversely, electrofishing is a sampling technique that actively pursues fish of a wide range of sizes. While many smaller age-0 fish can be easily overlooked during electrofishing, larger age-0 fish can be represented in samples, depending on the species. Electrofishing is effective for capturing sunfish, black bass, and shad (when present along the shoreline). Figure 1 illustrates the electrofishing catch rates of these species by reservoir. Electrofishing catch rates are based upon the mean of all sample events. In general, the representative reservoirs represent the high (Monticello Reservoir), mid (Fairfield Reservoir), and lower (Squaw Creek Reservoir) ranges of the abundance of these species. However, the variability in abundance appears to be wide from year to year, as illustrated for selected species in Figure 2. This variability can be the result of the combination of factors, such as reservoir water level changes (available habitat), variability in population sizes, increased electrofishing efficiency over time, and the change from fixed-station sampling to random sampling. Relative abundance of black bass, shad, and sunfish were consistently low in electrofishing samples taken at Squaw Creek Reservoir. This is somewhat surprising because anecdotal information indicate that when the reservoir was open to the public, it was considered one of the state's best bass fishing lakes. TPWD (Floyd Teat, personnel communication) suggests that electrofishing catch rates are low at Squaw Creek Reservoir due to the high conductivity, steep shorelines, and exceptionally clear water, which significantly decreased sampling efficiency. Thus, electrofishing samples might not be truly representative of the fish community.

Some additional generalities can be drawn from the data without detailed analysis. Since most of these reservoirs are not located on large rivers or streams, white bass (*Morone chrysops*) are either not present or occur in low numbers in these reservoirs (i.e., cooling ponds). The exceptions to this are lakes Eagle Mountain, Ray Hubbard, and Colorado City, which support appreciable numbers of white bass (Table 2). Eagle Mountain and Ray Hubbard reservoirs are large impoundments of rivers, which tend to maintain viable white bass populations. Striped bass (*M. saxatilis*) are absent or do not occur in large numbers in any of the reservoirs. Striped bass are not native to the areas where these reservoirs are located and only occur where they are stocked. Palmetto bass (*M. chrysops* x *M. saxatilis*), similar to striped bass, occur in a few of the reservoirs, but they only exist due to stocking. Since the reproductive potential of temperate bass for most of these reservoirs is very low to nonexistent, it would make sense to exclude these species as indicators. However, some consideration should be given to the reservoirs that support viable populations of white bass. Similar to the temperate bass, data indicate that crappie (*Pomoxis* spp.) are generally scarce in most of the reservoirs. The exceptions to this are Eagle Mountain, Graham, and Granbury reservoirs where appreciable numbers of crappie have been collected. In general, crappie do not thrive in reservoirs that were constructed as cooling ponds. TPWD attempted to establish crappie fisheries in many cooling ponds in the 1990s by introducing hybrid white (*P. annualis*) x black (*P. nigromaculatus*) crappie. However, their data indicated that the hybrids did not increase the crappie populations. The earlier cove rotenone data (see Table 3) also indicate that crappie populations were

historically low for most of the reservoirs. Anecdotal information from TXU staff indicate that anglers do catch crappie in some of the reservoirs, even though TPWD data indicates the populations are low.

Channel catfish (*Ictalurus punctatus*) are abundant in most of the reservoirs. Flathead catfish (*Pylodictis olivaris*) were considered; however, few reports contained data on this species. The data that were present indicated low abundance of this species across reservoirs. Blue catfish (*I. furcatus*) are generally absent from most of the reservoirs (see Table 2) because they are inhabitants of big rivers and are usually not found in small watersheds (Robison and Buchanan, 1992), unless they are stocked. Ray Hubbard is the only reservoir that maintains a large population of blue catfish. Even though channel catfish are abundant, the potential impingement rate of this species is questionable. Informal observations of intake screens at TXU and other facilities indicate that channel catfish impingement is typically low.

One detail that is not represented in the data is the impact that golden algae (*Prinnesium parvum*) have had on the fisheries in Colorado City and Granbury reservoirs. It would be difficult to determine quantitative effect on these reservoirs, but it is very possible that the recent sampling data do not represent the post fish-kill environments. However, the magnitude of the fish kills on Colorado City Reservoir was considered higher than the fish kills on Granbury Reservoir. In either situation, the actual numbers of fish susceptible to impingement might be low.

Based on available data, sunfish, shad, and black bass survey results might be the most appropriate data set for comparing reservoirs fish communities in relation to impingement. Compared with the inherent bias of capturing larger fish with gill nets and trap nets, electrofishing data likely represent the portion of the fish community vulnerable to impingement. This does not necessarily mean that the species targeted by gill nets and trap nets are not important to consider, but extrapolating the juveniles of those species would add a level of uncertainty. On the other hand, electrofishing data provide a direct measure of the relative abundance of target species. The catfish, temperate bass, and crappie data were compared between reservoirs, as discussed in the following section. However, these species were not used for developing habitat to fisheries relationships, as discussed in Section 3.2.3.

3.2.2 Analysis of Recent Data

CPUE data (electrofishing, gill netting, and trap netting) for various species were available from about 1986 to 2003 for 12 reservoirs (see Table 2). These data represent multiple years of data collection and the number of data points for the reservoirs ranged from 5 to 11. An analysis was performed to determine whether there is a statistically significant difference between the mean CPUEs of any pair of reservoirs. The analysis was performed for each of the indicator species.

The first step was to check whether the variance increased with the mean. Such condition would affect the validity of the results. The variance was stabilized with either logarithmic transformation or square-root transformation. The data were log-transformed when there was no zero in the data set; otherwise, the square-root transformation was used.

Then, analysis of variance (ANOVA) was performed for the data sets of all the reservoirs for a particular species. The null hypothesis was that the mean CPUEs of all the reservoirs were the same. If the ANOVA result supported the null hypothesis, then no further testing was necessary; otherwise, the next step was to determine which pair of reservoirs had different mean CPUEs. It was concluded that for each species, the null hypothesis was rejected at the 95 % confidence level.

The Tukey-Kramer method (Neter et al., 1990) was used for multiple comparisons of pairs of reservoirs. With this method, for each species, the pairs of reservoirs with different mean CPUEs at the 95% confidence level were identified.

Results of this analysis are shown in Table 5. There appears to be broad similarities in the indicator species between reservoirs. These results are in general agreement with the catch rates of black bass, sunfish, and shad illustrated in Figure 1. Fairfield Reservoir, which is considered "average" among the reservoirs, shares the most similarities in the fish community, followed by Monticello Reservoir and Squaw Creek Reservoir. If black bass, sunfish, and shad are used as a measure of similarity, as shown below, there are significant similarities between all of the reservoirs and at least one of the representative reservoirs (see Table 5). Reservoirs that are significantly similar based on two of the three species are denoted with an asterisk.

Fairfield	Monticello	Squaw Creek
Colorado City	Fairfield	North
Martin Creek	Tradinghouse Creek	Colorado City
Monticello	Martin Creek	Granbury
Striker	Eagle Mountain	Striker*
North	Ray Hubbard	Tradinghouse*
Tradinghouse Creek	Graham	Martin Creek*
Graham	Colorado City*	Graham*
Ray Hubbard	Striker*	Ray Hubbard*
Eagle Mountain*	Granbury*	
Granbury*		

3.2.3 Analysis of Older Data

Cove rotenone data were available for most reservoirs (Table 3). The species were grouped similarly to the above. The data were mostly from the mid 1980s and probably do not represent current conditions due to reservoir aging and habitat dynamics, changes in trophic state, and changes in fisheries management. However, these data provide the only estimates of standing crop, which could be one of the most meaningful measures for comparing reservoirs and estimating impingement. In theory, catch rates of electrofishing, gill netting, and trap netting should be positively correlated with standing crop. That is, reservoirs with more target fish should yield larger electrofishing, gill net, and trap net sample sizes. Unfortunately, there is little to no overlap between estimates of standing crop and the sample techniques

currently employed to determine whether this is indeed the case. However, the relationships between the latest available cove rotenone data and mean electrofishing CPUE for various species were investigated. As shown on Figure 3, there appears to be no relationship between recent electrofishing catch rates and previous standing crop. Figures 3c and 3d compare total numbers and standing crop with the electrofishing survey closest to that period. In two cases, the electrofishing surveys were conducted during the same year of the cove rotenone survey, but in most cases there were at least 2 years difference between sample events. Assuming enough reservoirs were concurrently sampled using both techniques, a relationship between electrofishing catch rate and standing crop might be expected. However, due to the wide disparities between survey events and the age of the data, it appears that the existing standing crop estimates cannot be confidently applied for comparisons of these reservoirs under current conditions.

3.2.4 Habitat Analysis

Comparisons of fish habitat can provide a measure of reservoir similarities, but would largely be an indirect measure of fish community characteristics. It has been well documented that physical habitats such as aquatic plants, flooded terrestrial vegetation, and rocks are an important life requisite for the sunfish, black bass crappie, and channel catfish. The importance of these variables are summarized in the Habitat Suitability Index Models developed by the U.S. Fish and Wildlife Service (FWS) (Robison and Buchanan, 1992; Miller, 1987; FWS, 1982a; FWS, 1982b; FWS, 1982c; FWS 1982d). As such, the quality and quantity of available habitats can be a good predictor of the type of fish community a reservoir can support.

Recent habitat data were available for all of the reservoirs open or recently open to the public. However, the methods for measuring and recording habitats vary between reservoirs. For instance, the number of habitat fields for reservoirs ranged from 4 to 23. In some instances, habitat surveys were confined to the shoreline, whereas in other reports, open-water habitats were documented separately. Some subjectivity was required to enable some type of comparative analysis. A suite of habitat categories was developed and are shown in Table 4. These categories were further distilled into even broader habitat categories, which include aquatic vegetation coverage, brush/trees, rock/gravel (including rip-rap), boat docks/piers, and non-descript. To help normalize data between reservoirs, the shoreline percent coverage and open-water percent coverage was averaged to develop a mean for each habitat category. The abundance of each of the habitats for each reservoir are shown on Figure 4. Squaw Creek Reservoir has the highest percentage of all habitats combined due to the high abundance of rocks/gravel and submerged trees/brush. Monticello Reservoir had the highest percent coverage of aquatic vegetation. However, up until about 2001, Martin Creek Reservoir probably had the highest percent of aquatic vegetation coverage due to the invasion of hydrilla (*Hydrilla verticillata*), which was recently, in part, eliminated by low water levels and the introduction of grass carp (*Ctenopharyngodon idella*).

The relationship between black bass, sunfish, and shad and each of the habitat categories, including a total of the habitats, was developed and is illustrated on Figure 5. A regression line was plotted and the

proportion of the variance is expressed as R^2 in each of the charts. A P-test was applied at the 95 % (0.05) level to determine whether the relationships were significant. In all cases but two, there were no significant relationships between the electrofishing catch rates and the habitat categories. However, the relationship between sunfish and black bass electrofishing catch rates and percent total aquatic vegetation appears to be significant. Shad catch rates were not expected to be correlated with the cover types, although the studies suggest some dependence of these species on inundated vegetation for spawning (FWS, 1985; Robison and Buchanan, 1992). Since shad feed primarily on plankton, abundance of these species is probably more closely related to the fertility (trophic state) of the reservoirs.

These results do not come as a big surprise. It is well known that the relationship between aquatic vegetation and sunfish (including black bass) is usually positively correlated. However, due to the amount of error introduced by variations in sampling techniques and by lumping the habitat types together, there is a possibility that these relationships are a false positives and some of the others might be false negatives. In addition, the habitat categories were based on a one-time sample event. It is possible that the habitat surveys at this time were not representative of the normal habitat conditions in the reservoir due to differences in water levels and weather. However, normalizing the habitat data was not possible because only limited habitat data were available. Statistical comparisons between each reservoir and habitat coverage was not possible since only one survey (data point) was available for most reservoirs.

What comes as a surprise was the slight negative relationship between rocks/gravel and electrofishing catch-rates for sunfish, although not significant. However, part of this is explained by the high numbers of sunfish in Monticello Reservoir where practically no gravel or rocks are present. Instead, sunfish abundance in that reservoir was probably more closely tied to the abundance of aquatic vegetation. Although sunfish catch rates in Squaw Creek Reservoir were very low this reservoir has the highest composition of rock and gravel. As previously mentioned, electrofishing data from this reservoir might not be truly reflective of the population due to the high conductivity clear water and steep slope.

The apparent lack of relationships is generally inconsistent with what is known about cover-dependent species. However, with the available data it is difficult to support grouping the reservoirs based on habitat similarities or dissimilarities. The exception to this is aquatic vegetation. When comparing black bass and sunfish populations in reservoirs, the data indicate that aquatic vegetation should be considered. The $\pm 25\%$ error range was arbitrarily selected since statistical analysis were not possible. If we take the similarities in aquatic vegetation coverage, while assuming this sample error, the following reservoirs can be grouped together.

Monticello	Fairfield	Squaw Creek
Tradinghouse	Tradinghouse	Eagle Mountain
	Ray Hubbard	Martin Creek
	North	Colorado City
	Striker	Granbury

Graham Reservoir is the only reservoir that did not fall into the $\pm 25\%$ range of any of the representative reservoir. However, this is somewhat of a broad grouping since the error range was arbitrarily selected. Due to the weaknesses of these comparisons and since habitats provide an indirect measure of fish populations, grouping reservoirs based on habitats should be secondary to grouping the reservoirs based on the fish communities.

3.2.5 Reservoir Physical Characteristics

Physical characteristics of this set of reservoirs vary widely based on their sizes, depths, stream order of the impounded stream, shoreline development ratio (SDR), and their location in relation to the state's vegetative regions (a characterization of the vegetative regions can be found in Gould [1962]). Reservoir size and their source waters probably account for some of the variations in fish populations, such as the abundance of white bass and crappie in the main-stem reservoirs. Fish species distribution can also vary from one region to the next, but with a few exceptions, that pattern generally applies to natural surface waters. Species composition probably varies with longitude in these reservoirs, but this type of analysis could not be performed due to scarcity of non-game species accounts. Instead, the data indicate that species composition between reservoirs is largely homogenous, at least for species targeted by fisheries management. In addition, catch rates of representative species across ecoregions and latitude were assessed, but did not demonstrate any type of correlation. This can be seen on Figure 2, where catch rates of the indicator species is independent of the location of the reservoir.

Catch rates of the indicator species were also independent of reservoir size. Although each analysis is not presented in this document, an example is presented on Figure 6a where sunfish catch rates were independent of reservoir size. Monticello Reservoir, which is one of the smallest reservoirs, had the highest catch rates for sunfish. This is interesting because if the impacts of impingement are high due to these facilities, this would mean that Monticello Reservoir, which supports a base-loaded facility and is one of the smallest reservoirs, should have low catch rates of the indicator species. Instead, it appears to be the most productive of all of the reservoirs. This issue was not addressed in this study, but there appears to be little relationship between the facility to volume ratio and catch rates of indicator species.

Similar to reservoir size, SDRs were also independent of catch rates of indicator species. SDR is the ratio between the amount of shoreline length to open water. A higher SDR means there is a higher amount of shoreline to open water (an SDR of 1 is a circle or has the least amount of shoreline possible). An example of sunfish catch rate compared with the SDRs is provided in Figure 6b. While reservoir size and SDR might be good indicators of a reservoir's ability to support a larger population sizes or standing crop, they essentially have no relationship with sampling catch rates. Larger reservoir sizes and higher SDRs suggest that more fish might be available for impingement; however, with increasing reservoir size and SDR, the potential interaction between CWISs and fish is decreased due to increased distance.

4.0 RECOMMENDATIONS FOR RESERVOIR GROUPING

Based on available fisheries data, there is good support for using the reservoirs identified by TXU to represent potential impingement at the remaining facilities. It will also be important to consider the facility operations, withdrawal rates, and CWIS characteristics for estimating impingement mortality at the remaining facilities. Since TPWD data do not exist for Trinidad or Valley Lake reservoirs, and only older, standing crop data exist for Lake Creek, some judgment regarding their fisheries will be required to identify an appropriate representative reservoir. Anecdotal information from TXU staff indicate that these reservoirs share similarities with Fairfield Reservoir. In turn, this information can probably be supported by the broad similarities Fairfield Reservoir shares with other reservoirs in this study.

The following table shows the reservoirs that can be grouped using similarities in fisheries as the primary criteria and aquatic vegetation as secondary criteria. The order in which the reservoirs are listed generally reflects the order of similarity. Reservoirs are listed multiple times since they share significant statistical similarities with more than one representative reservoir.

Fairfield	Monticello	Squaw Creek
Colorado City	Fairfield	North
Martin Creek	Tradinghouse Creek	Colorado City
Monticello	Martin Creek	Granbury
Striker	Eagle Mountain	Striker
North	Ray Hubbard	Tradinghouse Creek
Tradinghouse Creek	Graham	Martin Creek
Graham	Colorado City	Graham
Ray Hubbard	Striker	Ray Hubbard
Eagle Mountain	Granbury	
Granbury		

The analysis of available physical and biological data for the reservoirs considered in this study support grouping similar reservoirs together based on recently collected fisheries data. There were significant similarities between the catch rates of many of the common species targeted by fisheries managers and various reservoirs, which allowed for reasonable reservoir grouping by fisheries. From these data and anecdotal information about impingement in Texas reservoirs, it was concluded that sunfish and shad are among the species most common to these reservoirs and are likely to represent a majority of impingement numbers. The data also support that black bass are ubiquitous and relatively abundant, but this species is probably less prone to impingement because it remains of size susceptible to impingement only for a short period of its life. This is probably the case for many other species that attain larger sizes. As such, black bass, sunfish, and shad together are good indicators of reservoir health and are good candidates for measuring impingement.

Crappie, temperate bass, and catfish are also popular sport fish, but the sampling techniques preclude statistical analysis of the early life stages of these species, which would be the age group more susceptible to impingement. Catfish catch rates are among the highest in the representative reservoirs and would be well represented by these reservoirs. Only a few reservoir support substantial catch rates of crappie and temperate bass. Among these, Ray Hubbard Reservoir has relatively high catch rates of both groups of these fish. Since these species are not represented well in the representative reservoirs, it might be possible to conduct verification monitoring shortly after their reproductive season (late spring) to determine whether these species are susceptible to impingement. Considering the infrequent and unpredictable operation of this facility, sampling of this facility might not be practicable or possible for meeting quality assurance requirements.

Aside from the situation with crappie and white bass, measurements of habitats, locations, and physical characteristics of reservoirs appeared to have little bearing on catch rates of the indicator species. However, there was a positive relationship between aquatic vegetation and catch rates of sunfish and black bass, which is worth considering. The reservoirs that demonstrated a significant relationships based on aquatic plant abundance were in line with the reservoir grouping established based on the fisheries.

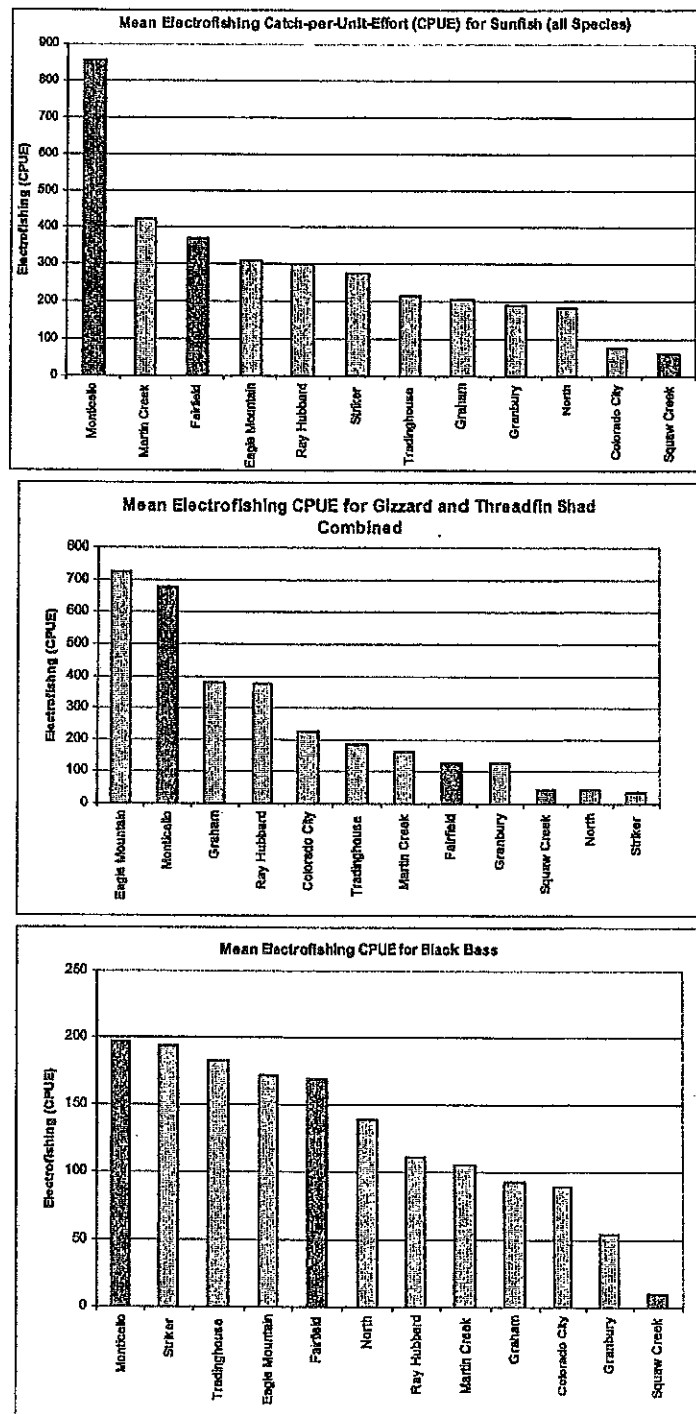
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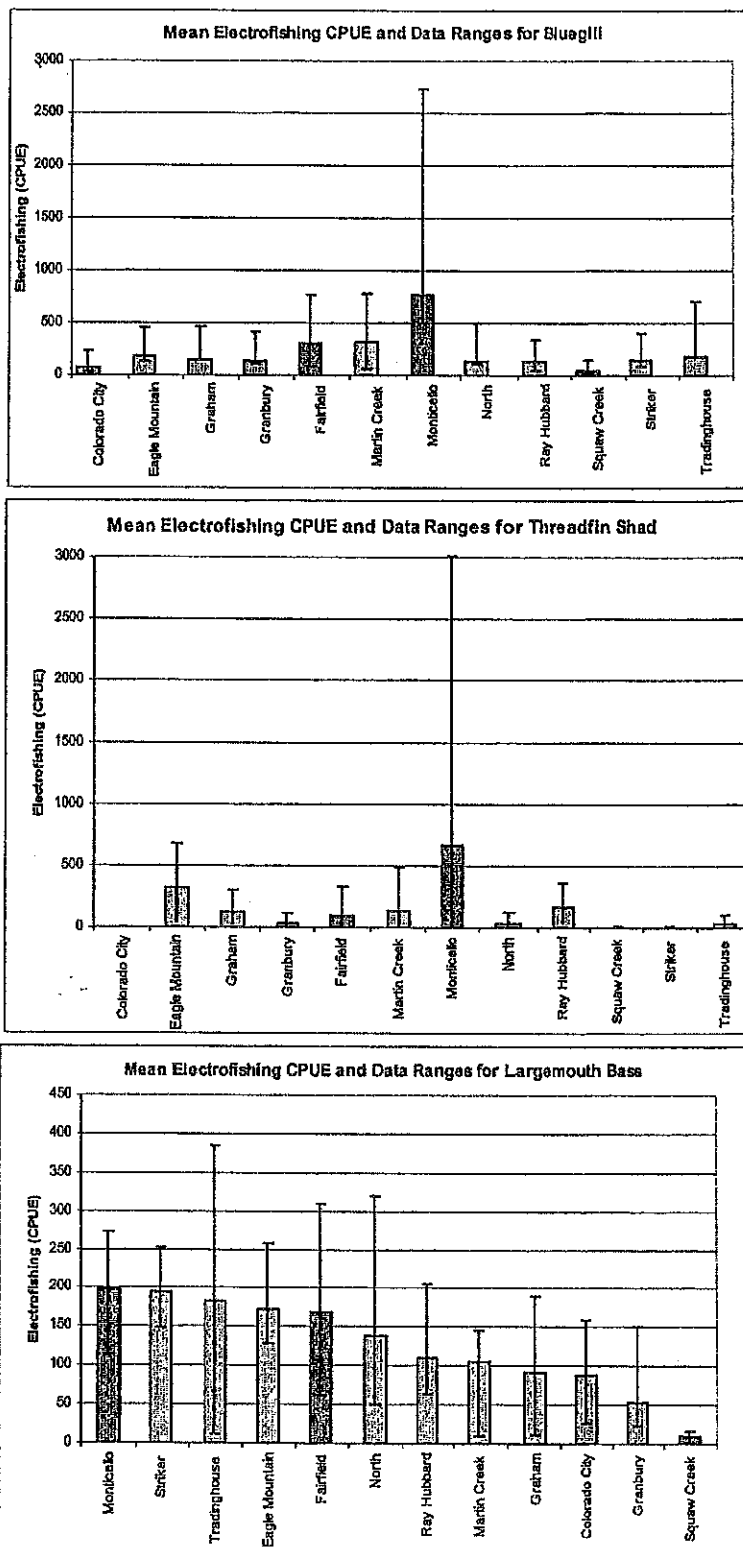
Figure 1: Mean Electrofishing CPUE for the Indicator Species, Sunfish, Shad, and Black Bass



Figures 1a, 1b, and 1c

Representative reservoirs are shaded in dark gray.
Source: TPWD performance reports.

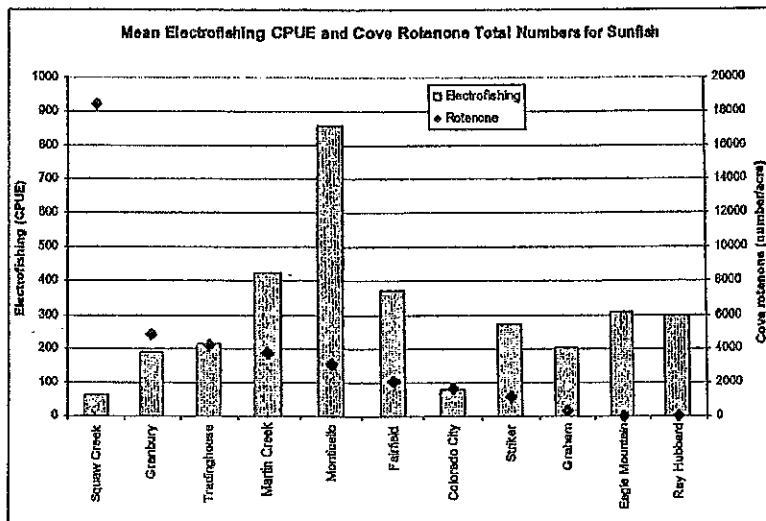
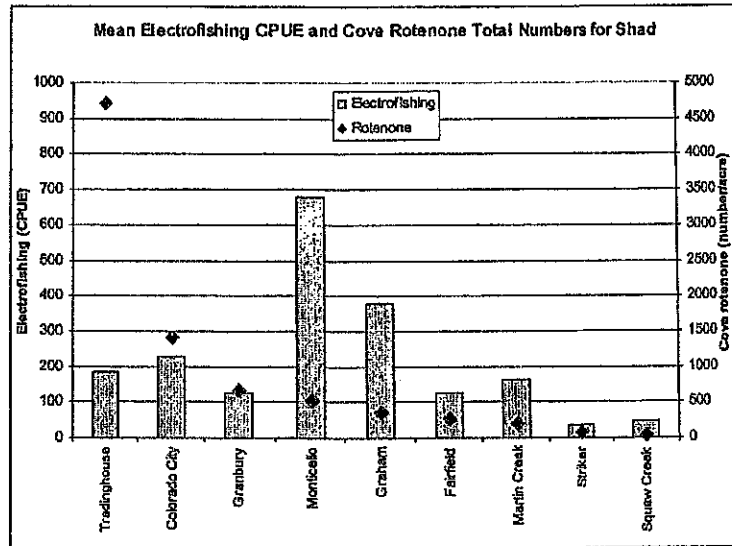
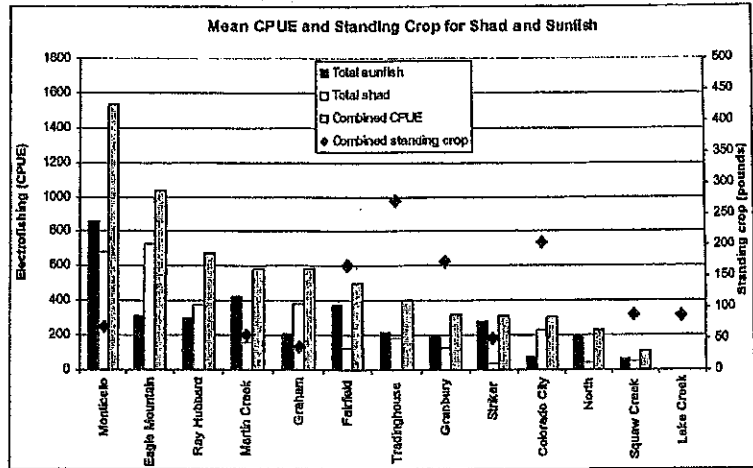
Figure 2: Mean Electrofishing and Data Ranges for Selected Species



Figures 2a, 2b, and 2c

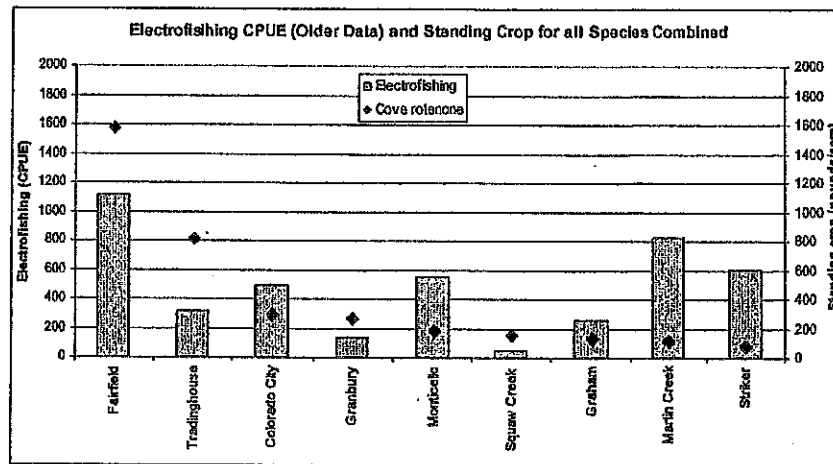
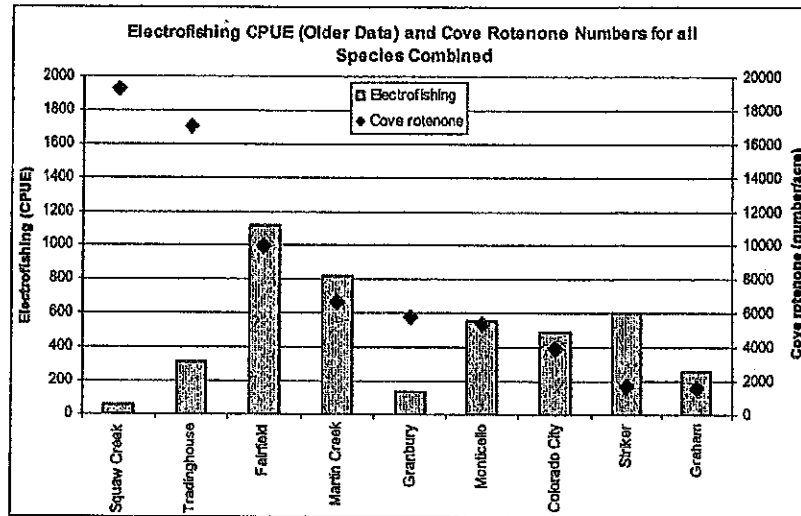
Error bars represent data ranges. Representative reservoirs are shaded in dark gray.
Source: TPWD performance reports.

Figure 3: Comparison Between Mean Electrofishing CPUE and Cove Rotenone Data for Various Species



Figures 3a, 3b, and 3c

Figure 3 (Cont'd)



Figures 3d and 3e

Source: TPWD performance reports.

Figure 4: Percent Coverage of Selected Habitats by Reservoir

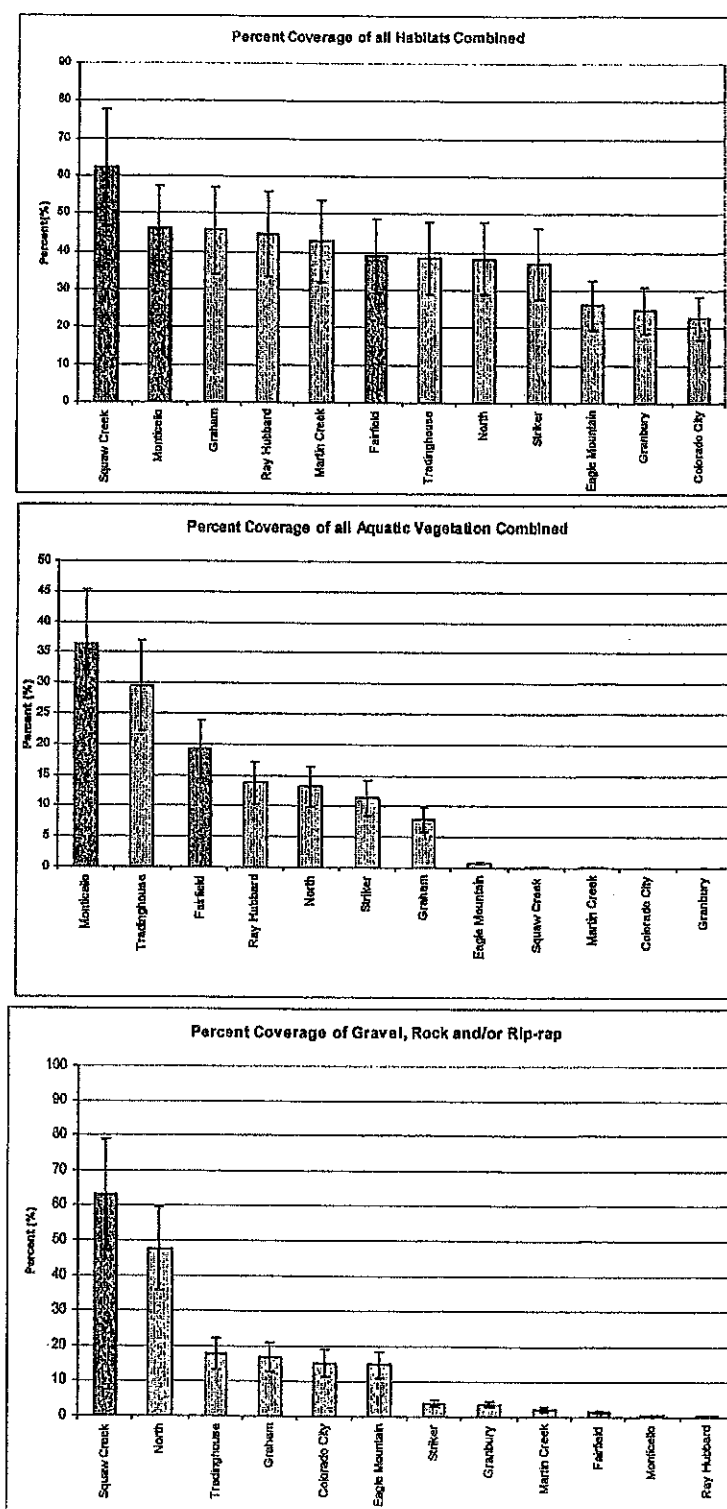


Figure 4a, 4b, and 4c

Figure 4 (Cont'd)

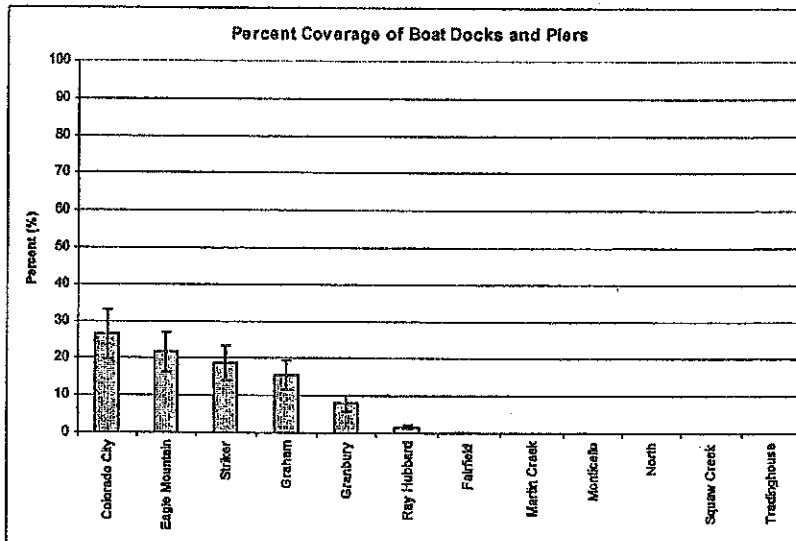
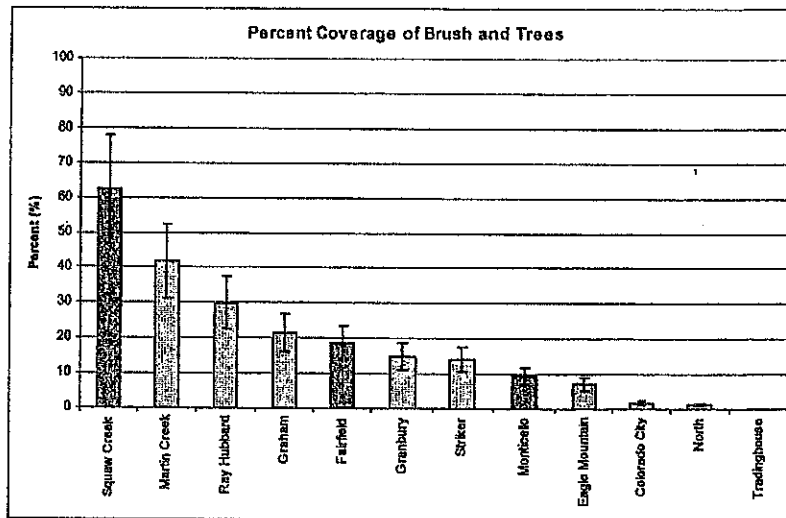
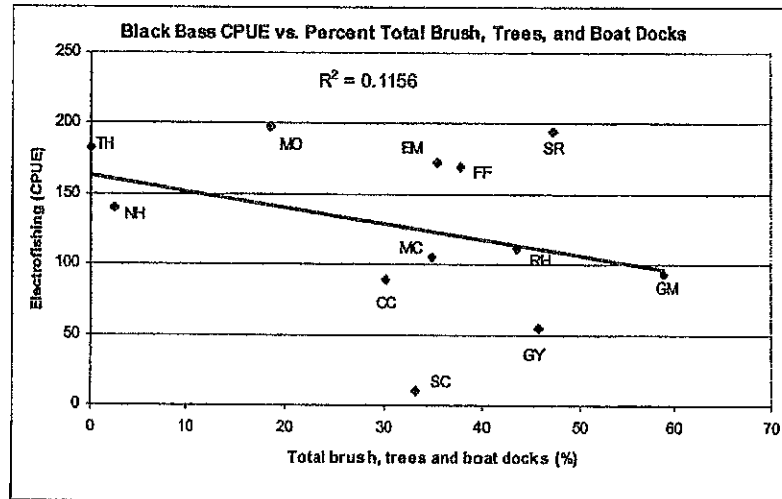
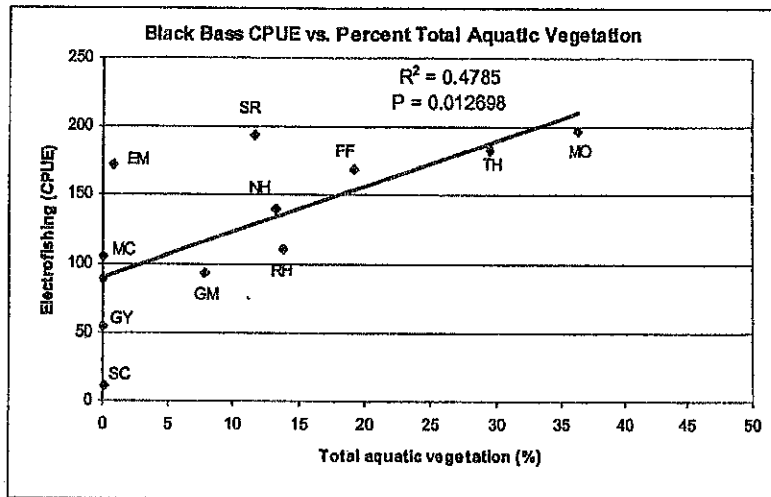
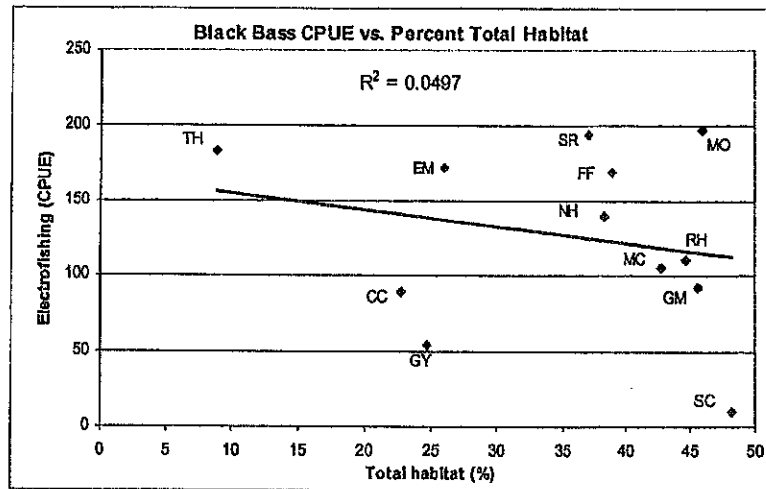


Figure 4d and 4e

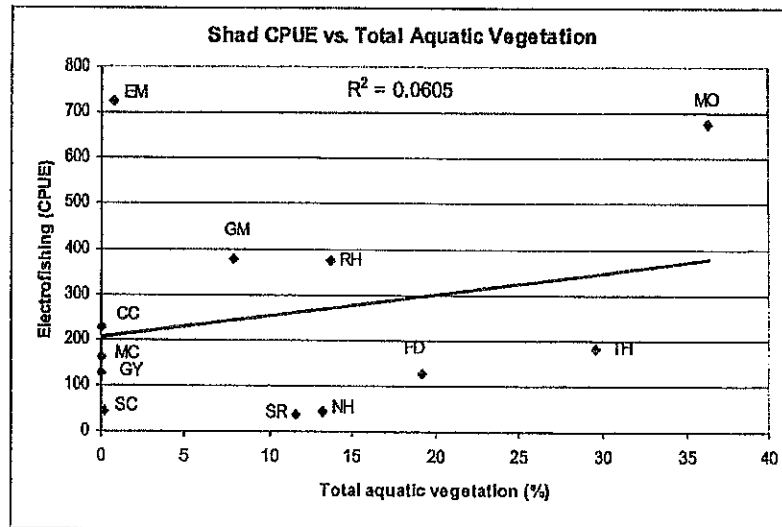
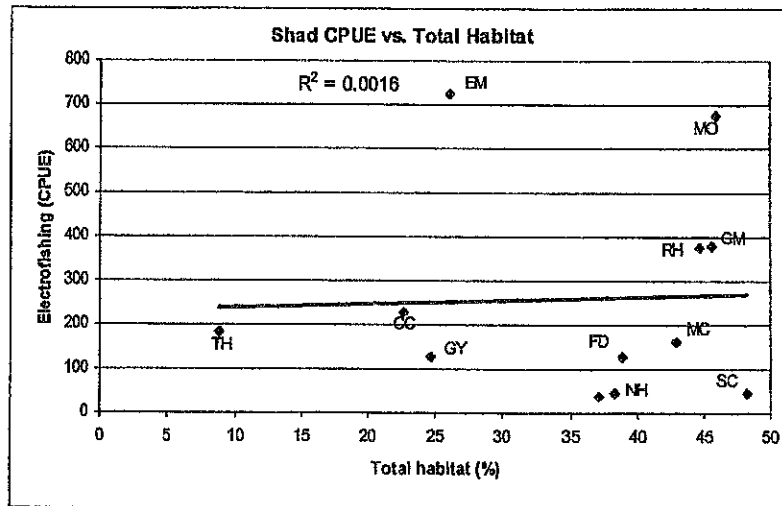
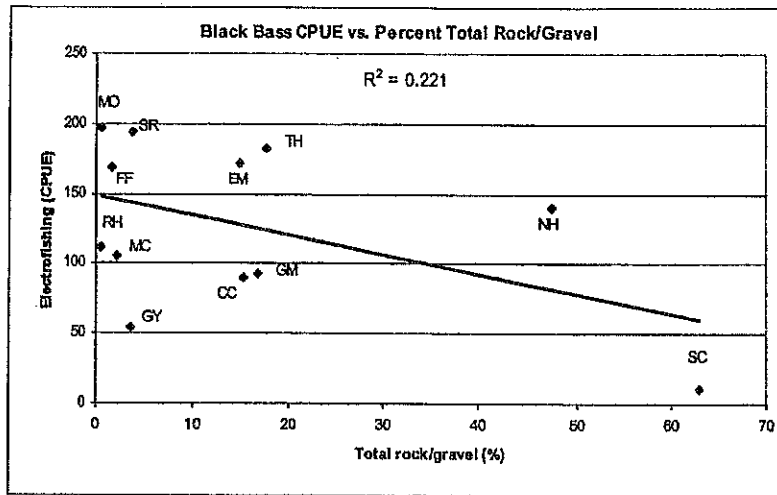
Error bars indicate $\pm 25\%$ error.
Source: TPWD performance reports.

Figure 5: Relationship Between Sampling Catch Rates of Selected Species and Various Habitat Types



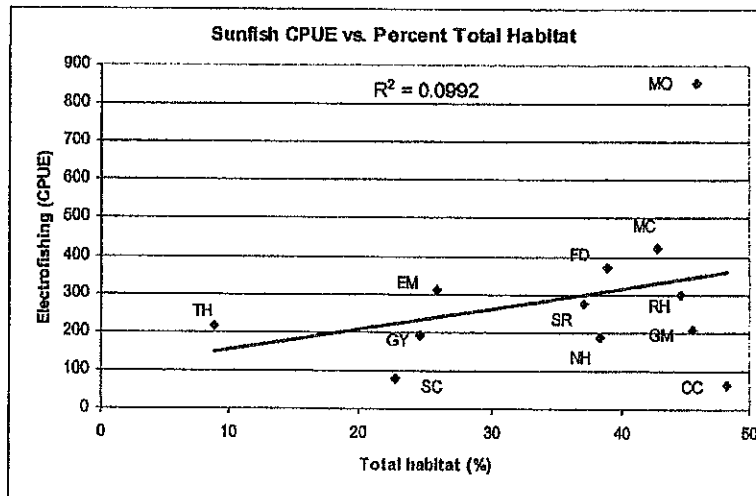
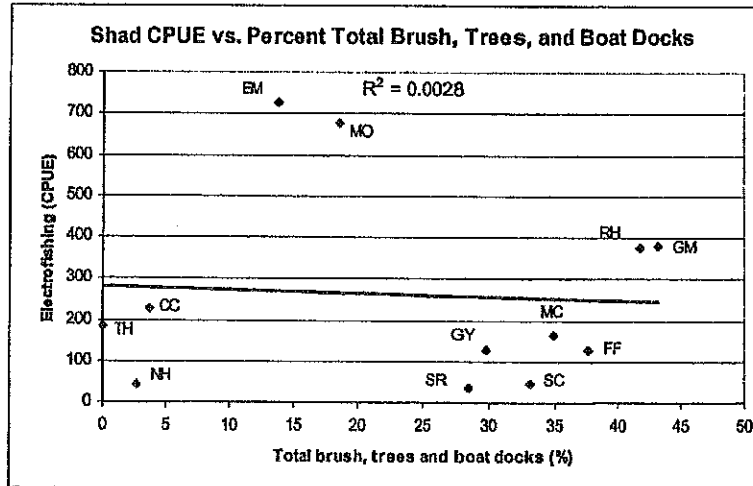
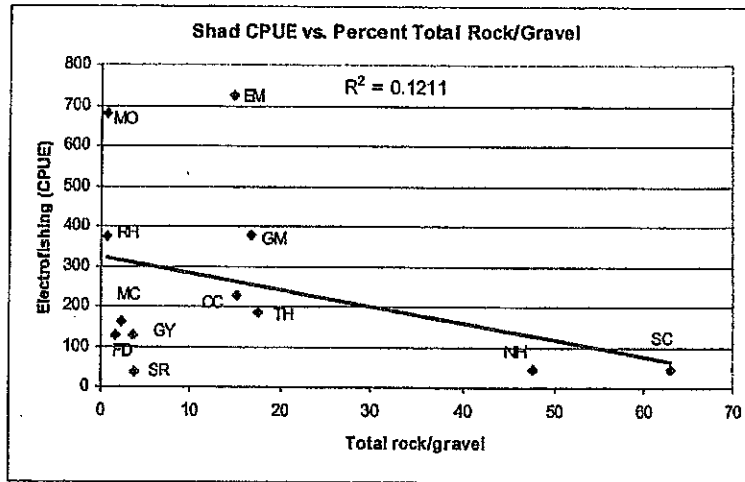
Figures 5a, 5b, and 5c

Figure 5 (Cont'd)



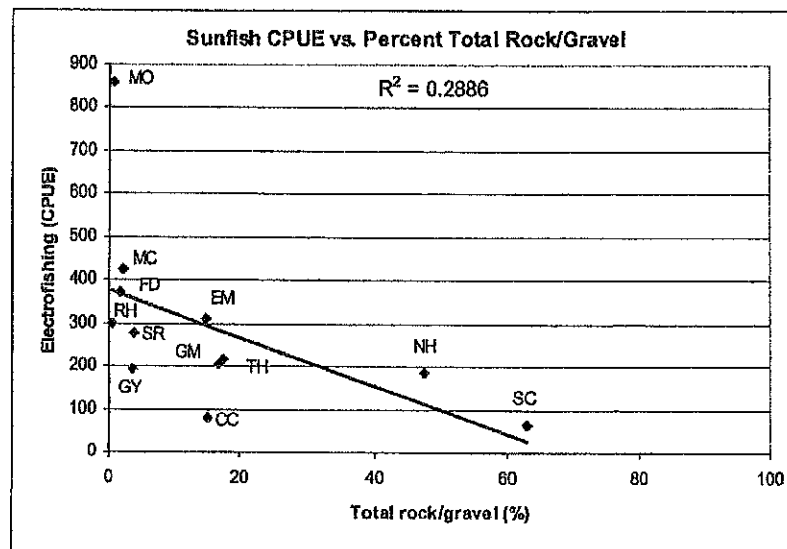
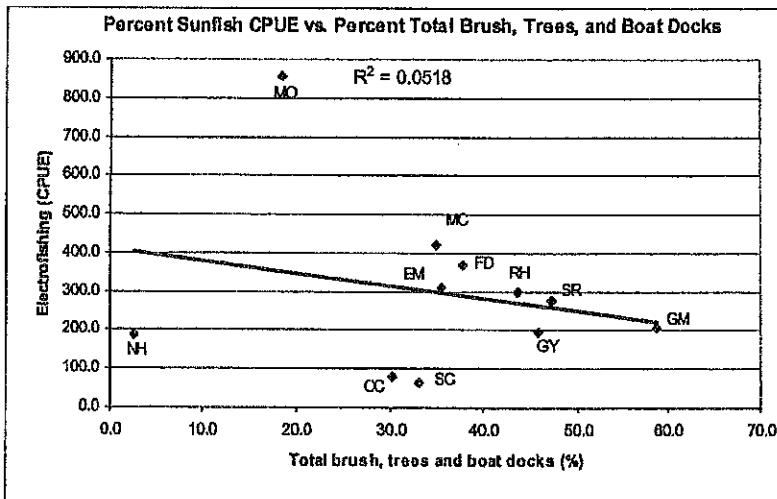
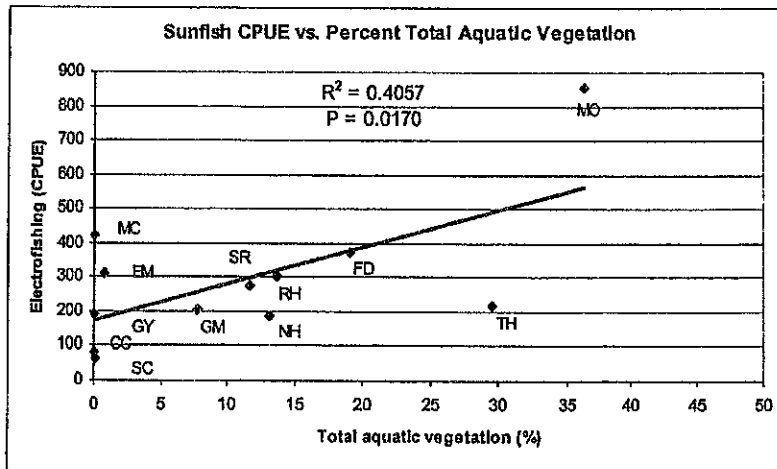
Figures 5d, 5e, and 5f

Figure 5 (Cont'd)



Figures 5g, 5h, and 5i

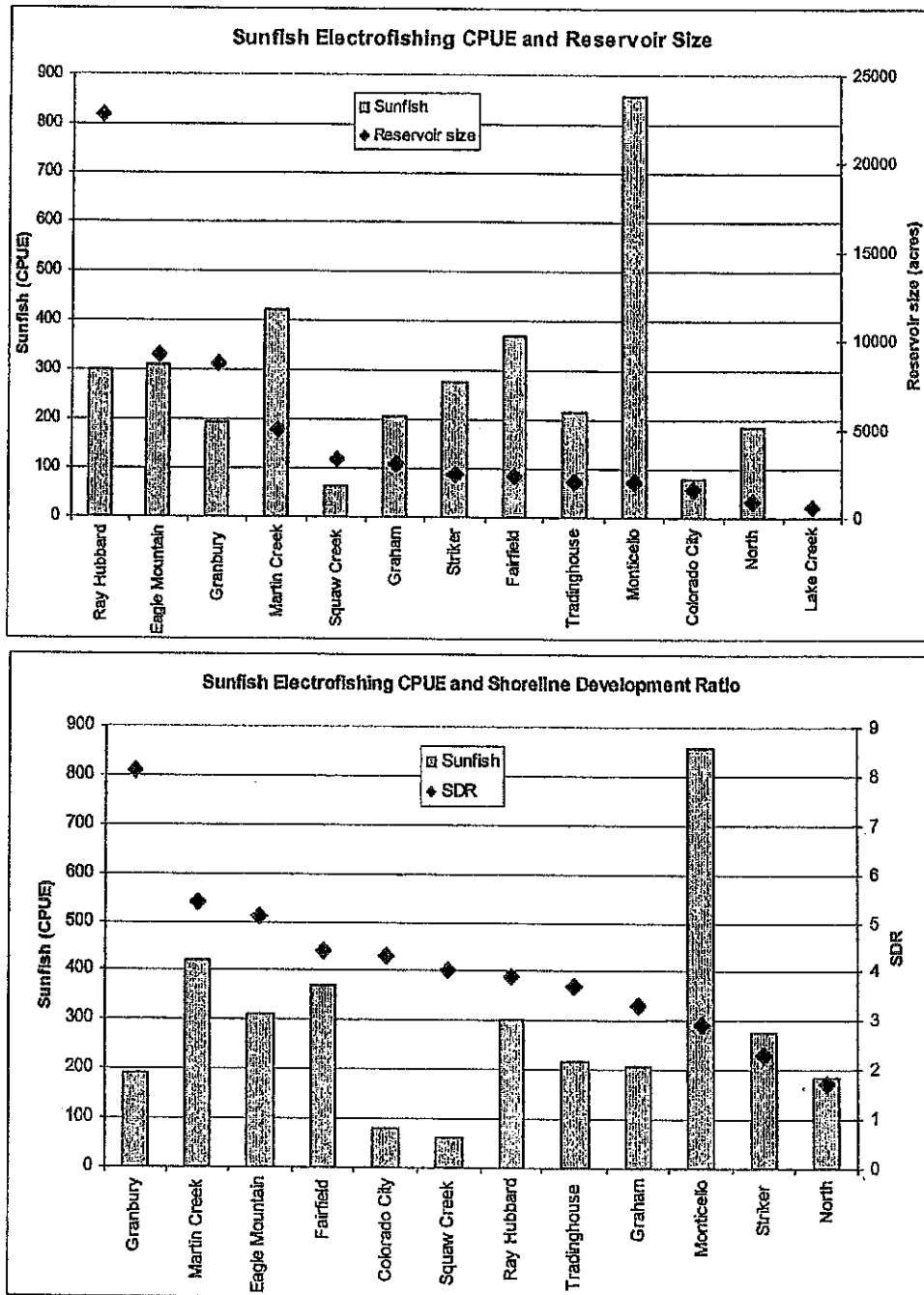
Figure 5 (Cont'd)



Figures 5j, 5k, and 5l

Source: TPWD performance reports.

Figure 6: Relationship Between Reservoir Sizes and Shoreline Development Ratios (SDR) and Sunfish Electrofishing Catch Rates



Figures 6a and 6b

Source: TPWD performance reports.

TABLE 1
RESERVOIR PHYSICAL CHARACTERISTICS

Reservoir	Report year	Size (acres)	Size (square feet)	Mean Secchi Depth (feet)	Mean Depth (feet)	Max Depth (feet)	Conductivity (mho/cm)	Shoreline Length (miles)	Shoreline Length (feet)	Shoreline Development Ratio	River System	Year Built	Controlling Authority	County (dam)
Colorado City	2001	1618	70480080	3.0	12.0	51.0	360	24.0	126720	4.3	Colorado	1949	TM	Michell
Eagle Mountain	2000	9200	400752000	3.0	22.0	47.0	230	69.0	364320	5.1	Trinity	1954	Tarrant County Water Control and Improvement District No.1	Tarrant
Graham	2001	3000	130680000	1.0	21.0	45.0	450	25.0	132000	3.3	Brazos	1929	City of Graham	Bug
Granbury	2001	8700	378972000	4.0	18.0	75.0	2400	106.0	559680	8.1	Brazos	1969	Brazos River Authority	Hood
Fairfield	2000	2353	102496680	5.0	21.5	50.0	850	30.0	158400	4.4	Trinity	1969	TM	Freestone
Lake Creek	1985	616	26832960	no data	14.4	35.0	no data	7.5	39487	2.2	Brazos	1951	TM	McLennan
Marlin Creek	2001	5000	217800000	2.5	16.0	50.0	120	53.0	279840	5.4	Sabine	1974	TM	Rusk
Monticello	1999	2800	87120000	3.8	20.3	40.0	335	18.0	95040	2.9	Cypress	1972	TM	Titus
North	1998	907	39508920	5.0	20.0	55.0	300	7.5	37725	1.7	Trinity	1957	TM	Dallas
Ray Hubbard	2001	22745	990772200	3.0	21.5	40.0	200	83.0	438240	3.9	Trinity	1968	City of Dallas	Rockwall
Squaw Creek	1997	3272	142528320	5.0	46.0	135.0	1300	32.0	168960	4.0	Brazos	1979	TM	Hood, Somerville
Striker	1999	2400	104544000	2.0	15.0	35.0	200	16.0	84480	2.3	Agelina		Agelina Macodoches County Water Control and Improvement District	Rusk
River Crest*														
Tradinghouse	2000	2012	87642720	3.0	19.0	42.0	1200	23.0	121440	3.7	Brazos	1968	TM	McLennan

Announced for retirement and will not be considered for 316(b) compliance.

Source: TPWD Reservoir performance reports.

TABLE 2
FISHERIES SURVEY DATA FOR SELECTED SPECIES BY RESERVOIR

Black Bass Electrofishing CPUE																			
Largemouth bass																			
Reservoir	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean
Colorado City			159.0			27.0			77.0	79.0		89.3				104.0			89.2
Eagle Mountain			173.0	121.0	110.0	119.0	222.7			150.0			93.3		105.3				136.8
Graham	23.3				12.0				90.0			149.3				188.8			92.7
Granbury			10.0		7.3		14.7			57.3			83.3			60.0			38.8
Fairfield		76.6	61.3			310.0		94.0			192.0			242.0	205.0				168.7
Martin Creek						120.7			154.0					146.0	9.4	96.7			105.4
Monticello				169.3	208.0	195.0	172.0	238.7	246.7	271.3	213.3	122.4	112.7	202.0					195.6
North			79.0			37.0	306.0			94.0			113.0						125.8
Ray Hubbard				88.0		63.0			101.0			206.5			96.0				110.9
Squaw Creek			10.7		2.0				12.7			16.0							10.4
Striker			127.3			70.7	55.3	56.0		43.3	51.3			80.0				24.0	69.1
Tradinghouse				59.3		11.1			384.0			181.0			277.0				182.5

Reservoir	Spotted bass																			Total	
	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean	Mean	
Colorado City			0.0			0.0			0.0	0.0		0.0				0.0			0.0	89.2	
Eagle Mountain			25.0		21.0		55.0			48.0			42.0		18.7				35.0	171.7	
Graham	0.0				0.0				0.0			0.0				0.0			0.0	92.7	
Granbury			6.7		4.0		8.0			14.7			35.3						15.5	54.3	
Fairfield		0.0	0.0			0.0		0.0			0.0			0.0	0.0				0.0	168.7	
Martin Creek						0.0			0.0					0.0	0.0	0.0			0.0	105.4	
Monticello				2.0	3.3	0.0	2.0	0.7	1.3	0.7	2.7	1.6	0.0	1.0					1.3	196.9	
North			6.0			6.0	16.0			27.0			5.0						13.5	139.3	
Ray Hubbard				0.0		0.0			0.0			0.0			0.0				0.0	110.9	
Squaw Creek			0.0		0.0					0.0		0.0							0.0	10.4	
Striker			108.7			111.3	132.7	171.3		72.7	258.0			77.0				66.0	124.7	193.8	
Tradinghouse												0.0			0.0				0.0	182.5	

Empty Cells indicate that there was no survey for that year. CPUE = Catch Per Unit Effort.
Source: TPWD Reservoir performance reports.

TABLE 2 (Cont'd)

Shad Electrofishing CPUE																			
Gizzard shad																			
Reservoir	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean
Colorado City			141.0			77.0			121.5			410.0			380.0				225.9
Eagle Mountain			214.0		291.0		329.0		274.0				589.3		711.8				401.5
Graham	48.7				84.7			261.3				202.7				694.0			258.3
Granbury			49.3		55.3		122.7		175.3				84.7			102.7			98.3
Fairfield		26.7	5.4			26.0		28.0			56.7			18.0	60.0				31.5
Martin Creek						43.3			14.7					55.3	11.3	12.7			27.5
Monticello				0.7	0.0	1.0	0.7	2.0	30.7	28.0	21.3	14.4	22.0	2.0					11.2
North			16.0			1.0	15.0		24.0				30.0						17.2
Ray Hubbard				163.0		78.0			309.0			293.5			177.5				205.4
Squaw Creek		10.0			6.0				7.3			152.0							43.8
Striker		24.7				26.7	25.3	32.0			30.0			38.0				64.0	34.4
Tradinghouse			32.7			4.0			148.0			121.0			441.0				149.3

Reservoir	Threadfin shad																			Total Mean
	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean	
Colorado City			0.0			0.0			0.0	115.3			0.0		3.0				0.6	226.5
Eagle Mountain			387.0		37.0		155.0						579.3		670.0				323.9	725.5
Graham	60.7				113.3			31.3		5.3		304.0				88.8			119.6	377.9
Granbury			0.0		110.0		36.0						4.0			14.7			28.3	126.7
Fairfield		25.3	332.0			100.0		6.0						21.0		89.0			95.6	127.1
Martin Creek						29.3			7.3					86.7	490.0	60.0			134.7	162.1
Monticello				239.3	615.3	1,040.0	1,007.3	3,010.0	143.3	267.3	401.3	230.4	5.3	370.0					666.3	677.5
North			4.0			0.0	2.0			1.0			126.0						26.6	43.8
Ray Hubbard				170.0		41.0			174.0			100.0			359.5				168.9	374.3
Squaw Creek			0.0		0.7				6.0				0.0						1.7	45.5
Striker			3.3			4.0	0.0	0.7						0.0			7.0		2.5	36.9
Tradinghouse				36.0		5.0			24.8			101.0			3.0				34.0	183.3

Empty Cells indicate that there was no survey for that year. CPUE = Catch Per Unit Effort.
Source: TPWD Reservoir performance reports.

TABLE 2 (Cont'd)

Catfish Gill Net CPUE																				
Reservoir		Blue catfish																		
		1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean
Colorado City				0.0			0.8			0.0			0.2				0.4			0.3
Eagle Mountain								0.0			0.7			0.6			0.9			0.6
Graham	0.0	0.0			0.0				0.0			0.0						0.7		0.1
Granbury								0.0			0.2			0.0			0.2	0.0		0.1
Fairfield			0.0				0.0		0.0			0.0			0.0		0.0			0.0
Martin Creek									0.0				0.2					0.0		0.1
Monticello				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0					0.0
North			0.0				0.0	0.0	0.0		0.0			0.0						0.0
Ray Hubbard				0.0			0.0			5.0			13.1				8.3			5.3
Squaw Creek			0.0			0.0				0.0			0.0							0.0
Striker			0.0			0.0	0.0	0.0	0.0			0.0			0.0				0.0	0.0
Tradinghouse				0.0			0.0		0.0	0.0		0.0			0.0	0.0	0.0			0.0

Channel Catfish																				
Reservoir		Channel Catfish																		
		1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean
Colorado City				9.0			2.8			1.6			1.2				1.6			3.2
Eagle Mountain				8.0		5.0		7.0			11.8			3.8			2.3			6.3
Graham	2.7	1.0			1.0					3.8			3.2					3.1		2.5
Granbury			5.0		3.7		7.2	4.4			4.9			3.9			5.5	5.0		4.6
Fairfield			15.5						0.8			2.6			7.0		17.4			8.4
Martin Creek							2.1			2.1			3.8					13.0		5.3
Monticello				43.4	20.2	25.8	22.8	14.2	14.2	24.4	0.0	20.6	14.6	20.2	13.8					20.0
North			2.0		0.0	2.0					1.4			2.6						1.6
Ray Hubbard				5.0		19.0				7.6			5.6				5.7			8.6
Squaw Creek			2.0		5.4					15.0			18.4							10.2
Striker			18.5			5.2			5.2			4.2			0.8				8.8	7.1
Tradinghouse				13.8		14.2				13.6		9.6			7.2	3.0	4.8			9.5

Empty Cells indicate that there was no survey for that year. CPUE = Catch Per Unit Effort.
Source: TPWD Reservoir performance reports.

TABLE 2 (Cont'd)

Temperate Bases Gill Net CPUE																				
White bass																				
Reservoir	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean	
Colorado City			7.8			0.8			1.4			16.8				30.0			11.4	
Eagle Mountain			7.0		9.0		3.0			10.3			11.1			3.7			7.4	
Graham		0.5	0.0		0.6				5.0			3.4					2.9		2.1	
Granbury			9.5		8.5		0.7			4.1			2.3			1.9	2.3		4.2	
Fairfield			0.0			0.0		0.0			0.0			0.0		0.0			0.0	
Martin Creek						0.0			0.0			0.0					0.0		0.0	
Monticello				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0					0.0	
North			3.0			0.0	4.0			3.2			0.6						2.2	
Ray Hubbard				9.0		12.0			11.3			5.7				3.5			8.3	
Squaw Creek			0.8		5.0				0.8			0.2							1.7	
Striker			7.3			0.6		0.0		0.0	0.2			0.0				3.0	1.6	
Tradinghouse				0.0		1.0			1.0			1.2		0.0	0.9	1.0			0.7	

Reservoir	Striped bass																			
	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean	
Colorado City			0.0			0.0			0.0			0.0				0.0			0.0	
Eagle Mountain			0.0		0.0		0.0			0.0			0.0			0.0			0.0	
Graham			0.0		0.0				0.0			0.0				0.0			0.0	
Granbury			2.0		0.7		2.3			1.9			1.7			0.4	1.9		1.6	
Fairfield			0.0			0.0		0.0											0.0	
Martin Creek						0.0			0.0			0.0					0.0		0.0	
Monticello				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0					0.0	
North			0.0			0.0	0.0			0.0			0.0						0.0	
Ray Hubbard				0.0		0.0				0.0		0.1				0.0			0.0	
Squaw Creek					0.0				0.0			4.8							1.2	
Striker			0.0			0.0		0.0		0.0	0.0			0.0				0.0	0.0	
Tradinghouse				0.0		0.0			0.0			0.0		0.0	0.0	0.0			0.0	

Reservoir	Palmetto bass (hybrid white male x striped female bass)																				Total Mean
	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean		
Colorado City			0.0			0.0			0.0			0.0				0.0			0.0	11.4	
Eagle Mountain			0.0		0.0		0.0			0.0		0.0				0.0			0.0	7.4	
Graham	0.2	0.2			4.6				9.0			10.0					2.7		4.5	6.5	
Granbury			0.0		0.0		0.0			0.0			0.0			0.0	0.0		0.0	5.7	
Fairfield			11.2			8.2		4.6				11.4		1.2		1.6		0.0	6.4	6.4	
Martin Creek						0.0			0.0			0.0						0.0	0.0	0.0	
Monticello				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				0.0	0.0	0.0	
North			0.0			0.0	0.0			0.0			0.0						0.0	2.2	
Ray Hubbard			10.0			2.0			7.9			5.1				0.0			5.0	13.3	
Squaw Creek			1.8		0.0				0.0			4.8							1.7	4.6	
Striker			0.0			0.0		0.0		0.0	0.0			0.0				0.0	0.0	1.6	
Tradinghouse				0.0		0.0			0.0			0.0		0.0	0.0	0.0			0.0	0.7	

Empty Cells indicate that there was no survey for that year. CPUE = Catch Per Unit Effort.
Source: TPWD Reservoir performance reports.

TABLE 2 (Cont'd)

		Crappie Trap Net CPUE																		
		White crappie																		
Reservoir		1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean
Colorado City				0.3			0.1			0.6			0.2	0.2		4.0				0.9
Eagle Mountain				4.0		9.0		5.0			1.4			3.4		3.2				4.3
Graham						11.0				5.6			10.8				3.6			7.8
Granbury				1.0		5.9		3.7		22.9				9.3			3.3			7.7
Fairfield			0.0	0.0			0.0		0.2		0.0				0.0	0.0				0.0
Martin Creek							1.1			2.7			0.1				0.0			1.0
Monticello					0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0					0.0
North								1.0			2.2			4.2						2.5
Ray Hubbard					5.0		7.0			9.0			6.1			14.1				8.2
Squaw Creek				0.0		1.0				0.0			0.2							0.3
Striker				1.6			1.8		1.5		1.9				0.2					1.5
Tradinghouse						1.6	0.8			4.2			0.8			1.8			1.8	1.3

		Black crappie																		
		Black crappie																		
Reservoir		1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	Mean
Colorado City				0.0			0.0			0.0			0.0	0.0		0.0				0.9
Eagle Mountain				0.0		0.0		0.0			0.5		0.0	6.4		0.7				1.3
Graham						0.2				0.0			0.0				0.0			5.6
Granbury				0.0		0.1		0.0			0.0			0.0			0.0			7.8
Fairfield		4.4	2.8				0.0		0.6			0.0		0.0			0.0			7.7
Martin Creek							1.6					0.0	1.2		0.0	0.0				1.1
Monticello					0.0	0.2	0.0	0.8	0.0	0.0	0.2	0.6	1.6	1.2	0.3		1.7			1.5
North								0.0			0.0			0.0						2.5
Ray Hubbard					0.0		0.0			0.0			5.3			2.3				0.4
Squaw Creek				0.0		0.0				0.0			0.2							1.5
Striker			0.4				0.2		0.0			0.0	0.2		0.0					9.8
Tradinghouse						0.0	0.0			0.0			0.0			9.8			0.2	2.0
																				3.8

TABLE 3
STANDING CROP AND TOTAL NUMBERS FROM COVE ROTENONE SURVEYS

Reservoir	Colorado City	Eagle Mountain	Graham	Granbury	Fairfield	Lake Creek	Martin Creek	Monticello	North	Ray Hubbard	Squaw Creek	Striker	Tradinghouse
Date	1985	No data	1983	1981	1985	1984	1985	1987	No data	No data	1981	1975	1987
Area (acres)	3	available	3	3.13	2.2	3	3.8	2.2	available	available	1.53	3.61	2
Standing Crop (pounds):													
Gizzard shad (<i>Dorosoma cepedianum</i>)	154.0		27.1	51.7	32.6	47.0	9.2	0.0			24.5	11.2	128.6
Threadfin shad (<i>D. petenense</i>)	3.1		1.3	4.2	0.8	1.8	1.9	5.7			0.0	0.1	26.1
Total shad	157.1	*	28.4	55.9	33.4	48.8	11.1	5.7	*	*	24.5	11.3	154.7
Largemouth bass (<i>Micropterus salmoides</i>)	13.1		4.2	6.6	144.1	16.0	11.5	53.8			47.0	14.5	69.0
Spotted bass (<i>M. punctulatus</i>)	0.0		0.0	0.8	0.0	1.4	0.0	0.0			0.0	2.3	0.0
Total black bass	13.1	*	4.2	7.4	144.1	17.4	11.5	53.8	*	*	47.0	16.9	69.0
White crappie (<i>Pomoxis annularis</i>)	0.0		1.8	2.5	0.0	0.3	1.0	0.0			0.0	0.0	0.2
Black crappie (<i>P. nigromaculatus</i>)	0.0		0.0	0.0	0.0	0.0	0.1	0.0			0.0	0.0	0.0
Total crappie	0.0	*	1.8	2.5	0.0	0.3	1.1	0.0	*	*	0.0	0.0	0.2
Channel catfish (<i>Ictalurus punctatus</i>)	13.2		4.7	8.1	36.4	6.3	2.0	55.1			1.9	3.3	234.7
Blue catfish (<i>I. furcatus</i>)	0.0		0.0	0.0	0.0	0.0	0.0	0.0			0.0	0.0	0.0
Total catfish	13.2	*	4.7	8.1	36.4	6.3	2.0	55.1	*	*	1.9	3.3	234.7
Bluegill (<i>Lepomis macrochirus</i>)	28.0		4.0	72.4	119.3	26.0	17.9	45.8			37.8	3.9	103.8
Other sunfish (<i>L. spp.</i>)	19.3		3.2	45.6	13.9	12.5	26.4	17.8			26.7	34.6	12.7
Total sunfish	47.3	*	7.3	118.0	133.2	38.5	44.3	63.6	*	*	64.5	38.5	116.5
White bass (<i>Morone chrysops</i>)	0.0		0.1	0.0	0.0	0.0	0.0	0.0			0.0	0.4	0.0
Striped bass (<i>M. saxatilis</i>)	0.0		0.0	0.0	0.0	0.0	0.0	0.0			0.0	0.0	0.0
Palmetto bass (<i>M. chrysops</i> x <i>M. saxatilis</i>)	0.0		0.0	0.0	0.0	0.0	0.3	0.0			0.0	0.0	0.0
Total temperate basses	0.0	*	0.1	0.0	0.0	0.0	0.3	0.0	*	*	0.0	0.4	0.0
Tilapia (<i>Oreochromis aureus</i>)	3.3			0.0	954.0	0.0	0.3	0.0			0.0	0.0	75.5
Other	52.3		83.3	71.8	275.0	31.7	41.3	4.4			16.3	9.2	159.4
Total	286.3	*	129.6	263.7	1,576.1	143.0	111.9	182.6	*	*	154.2	79.6	810.0

TABLE 3 (Cont'd)

Reservoir	Colorado City	Eagle Mountain	Graham	Granbury	Fairfield	Lake Creek	Martin Creek	Monticello	North	Ray Hubbard	Squaw Creek	Striker	Tradinghouse
Date													
Area (acres)													
Total number:													
Gizzard shad (<i>Dorosoma cepedianum</i>)	720.0		138.3	325.0	108.1	159.0	15.5	0.0			30.0	39.6	645.0
Threadfin shad (<i>D. petenense</i>)	679.0		220.4	343.0	155.2	662.0	183.4	518.1			0.0	32.5	4,075.0
Total shad	1,399.0	*	358.7	668.0	263.3	821.0	198.9	518.1	*	*	30.0	72.2	4,720.0
Largemouth bass (<i>Micropterus salmoides</i>)	142.5		146.1	73.0	1,817.7	114.0	404.2	1,473.2			340.0	109.2	376.0
Spotted bass (<i>M. punctulatus</i>)	0.0		0.0	16.0	0.0	21.0	0.0	0.0			0.0	19.8	0.0
Total black bass	142.5	*	146.1	89.0	1,817.7	135.0	404.2	1,473.2	*	*	340.0	129.1	376.0
White crappie (<i>Pomoxis annularis</i>)			13.2	49.0	0.0	9.0	2.8	0.0			1.0	10.4	1.0
Black crappie (<i>P. nigromaculatus</i>)			0.0	0.0	0.0	0.0	1.1	0.0			0.0	0.8	0.0
Total crappie	0.0	*	13.2	49.0	0.0	0.0	3.9	0.0	*	*	1.0	11.1	1.0
Channel catfish (<i>Ictalurus punctatus</i>)	110.0		53.3	17.0	68.9	34.0	21.7	64.4			2.0	51.3	690.0
Blue catfish (<i>I. furcatus</i>)	0.0		0.0	0.0	0.0	0.0	0.0	0.0			0.0	0.0	0.0
Total catfish	110.0	*	53.3	17.0	68.9	34.0	21.7	64.4	*	*	2.0	51.3	690.0
Bluegill (<i>Lepomis macrochirus</i>)	879.0		195.0	3,218.0	1,853.4	906.0	2,496.6	2,352.3			14,473.0	167.7	3,968.0
Other sunfish (<i>L. spp.</i>)	796.9		141.7	1,661.0	245.3	285.0	1,253.9	739.3			3,914.0	1,014.4	301.0
Total sunfish	1,675.9	*	336.7	4,879.0	2,098.7	1,191.0	3,750.5	3,091.6	*	*	18,387.0	1,182.1	4,269.0
White bass (<i>Morone chrysops</i>)	0.0		0.6	0.0	0.0	0.0	0.0	0.0			0.0	1.4	0.0
Striped bass (<i>M. saxatilis</i>)	0.0		0.0	0.0	0.0	0.0	0.0	0.0			0.0	0.0	0.0
Palmetto bass (<i>M. chrysops</i> x <i>M. saxatilis</i>)	0.0		0.0	0.0	0.0	0.0	0.0	0.0			0.0	0.0	0.0
Total temperate basses	0.0	*	0.6	0.0	0.0	0.0	0.0	0.0	*	*	0.0	1.4	0.0
Tilapia (<i>Oreochromis aureus</i>)	4.2		0.0	0.0	4,388.5	0.0	0.6	0.0			0.0	0.0	3,212.0
Other	611.6		737.5	54.0	1,284.8	456.0	2,280.0	244.9			521.0	254.1	3,776.0
Total	3,943.2	*	1,646.1	5,756.0	9,921.9	2,637.0	6,661.8	5,392.2	*	*	19,281.0	1,701.3	17,044.0

* Data not available

The most recent cover rotenone survey for each reservoir are represented above.

Source: TPWD Reservoir performance reports.

TABLE 4

ND = Data not available.
Habitats represent most current surveys for each reservoir. (Some habitat fields in the reservoir performance reports were combined.)
Source: TPWD Reservoir performance reports.

TABLE 5
FISH COMMUNITY COMPARISON BETWEEN RESERVOIRS

	Squaw Creek	Granbury	Eagle Mountain	Ray Hubbard	Graham	Striker	North	Monticello	Martin Creek	Tradinghouse	Colorado City	Fairfield	
Black bass													
Shad													
Sunfish													
Catfish													
Crappie													
Temperate bass													
	Squaw Creek	Granbury	North	Striker	Graham	Colorado City	Ray Hubbard	Eagle Mountain	Martin Creek	Tradinghouse	Fairfield	Monticello	
Black bass													
Shad													
Sunfish													
Catfish													
Crappie													
Temperate bass													
	Monticello	Eagle Mountain	Fairfield	Ray Hubbard	Graham	Martin Creek	Tradinghouse	Striker	Granbury	Colorado City	North	Squaw Creek	
Black bass													
Shad													
Sunfish													
Catfish													
Crappie													
Temperate bass													

Cells shaded in gray are the representative reservoirs. Cells without shading are indicate similarity at the 95% confidence level between reservoirs. Cells shaded in black indicate that the reservoirs are dissimilar.

Trinidad Steam Electric Station
WQ0000947000
PLAIN LANGUAGE SUMMARY

The following summary is provided for this pending water quality permit application being reviewed by the Texas Commission on Environmental Quality as required by 30 Texas Administrative Code Chapter 39. The information provided in this summary may change during the technical review of the application and are not federal enforceable representations of the permit application.

Luminant Generation Company LLC (CN603256413) operates the Trinidad Steam Electric Station (RN101943868), located at 1320 McEntire Rd., Trinidad, Henderson County, Texas 75163. The facility currently consists of one natural gas-fired steam electric generating unit.

This application is for the renewal of Texas Pollutant Discharge Elimination System (TPDES) Permit No. WQ0000947000 (EPA I.D. No. TX001031) which authorizes the discharge of wastewaters (once-through cooling and auxiliary cooling water) and previously monitored effluent (floor/equipment drains, equipment blowdown, water treatment wastes, low volume waste, metal cleaning waste and storm water) at a daily average flow not to exceed 425 million gallons per day via Outfall 001.

The discharge of once-through cooling water via Outfall 001 and previously monitored effluent via Outfalls 101, 201 and 301 from this facility is subject to federal effluent limitation guidelines at 40 CFR Part 423. The pollutants expected from these discharges based on 40 CFR Part 423 are: total residual chlorine, free available chlorine, total suspended solids, oil and grease, total iron, total copper and pH. Temperature is also expected from discharges of 001.

The raw water supply for the facility's cooling water and service water systems is from Trinidad Reservoir. A chemical feed system supplies water conditioning chemicals to the once-through cooling water to minimize corrosion and control the formation of mineral scale and bio-fouling. Domestic wastes are routed to an on-site septic system.